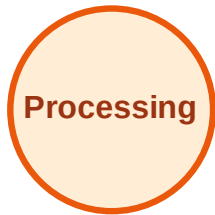


BFS Traversal

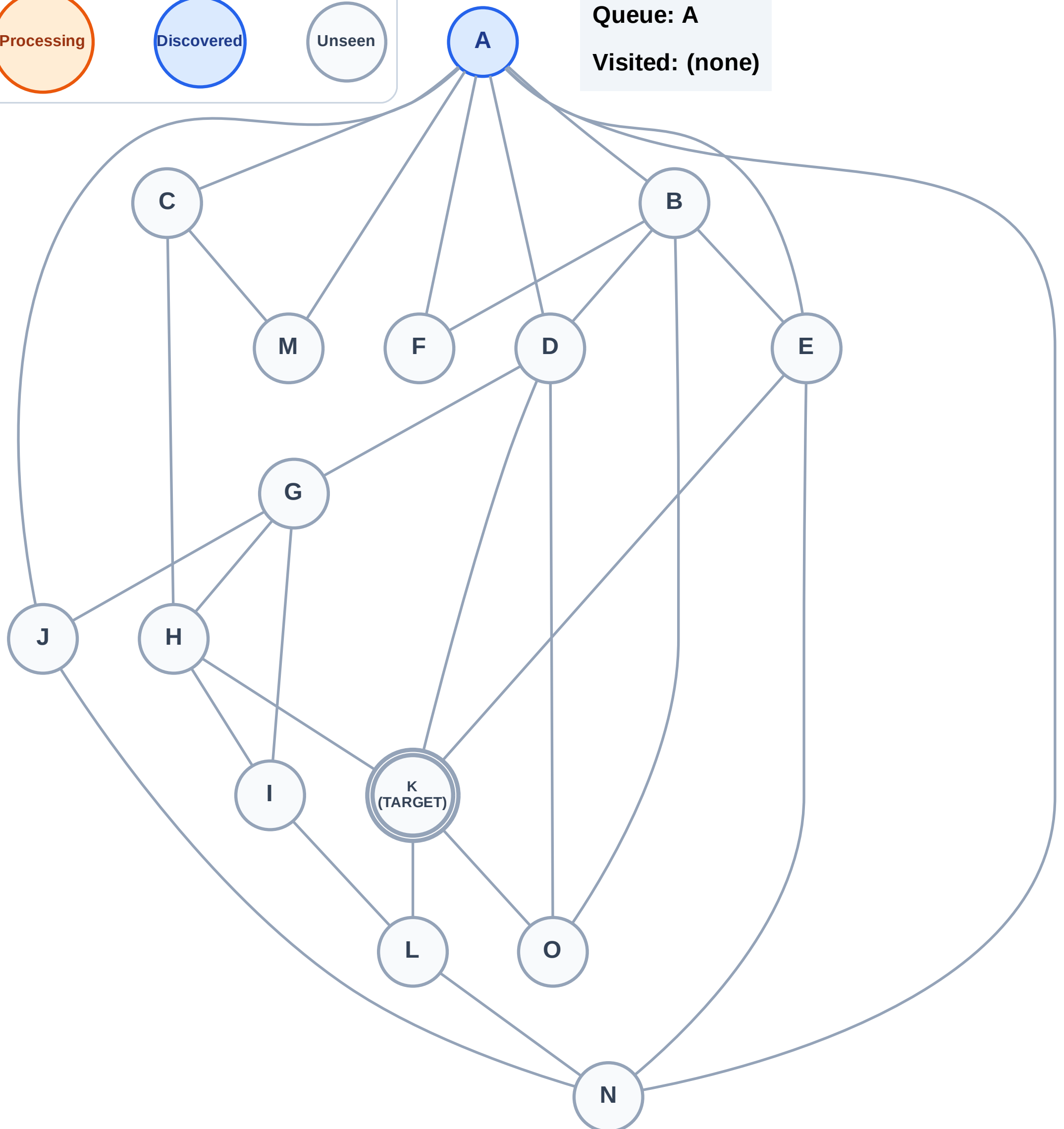
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



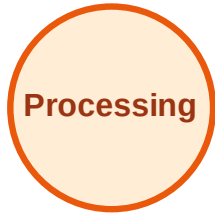
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



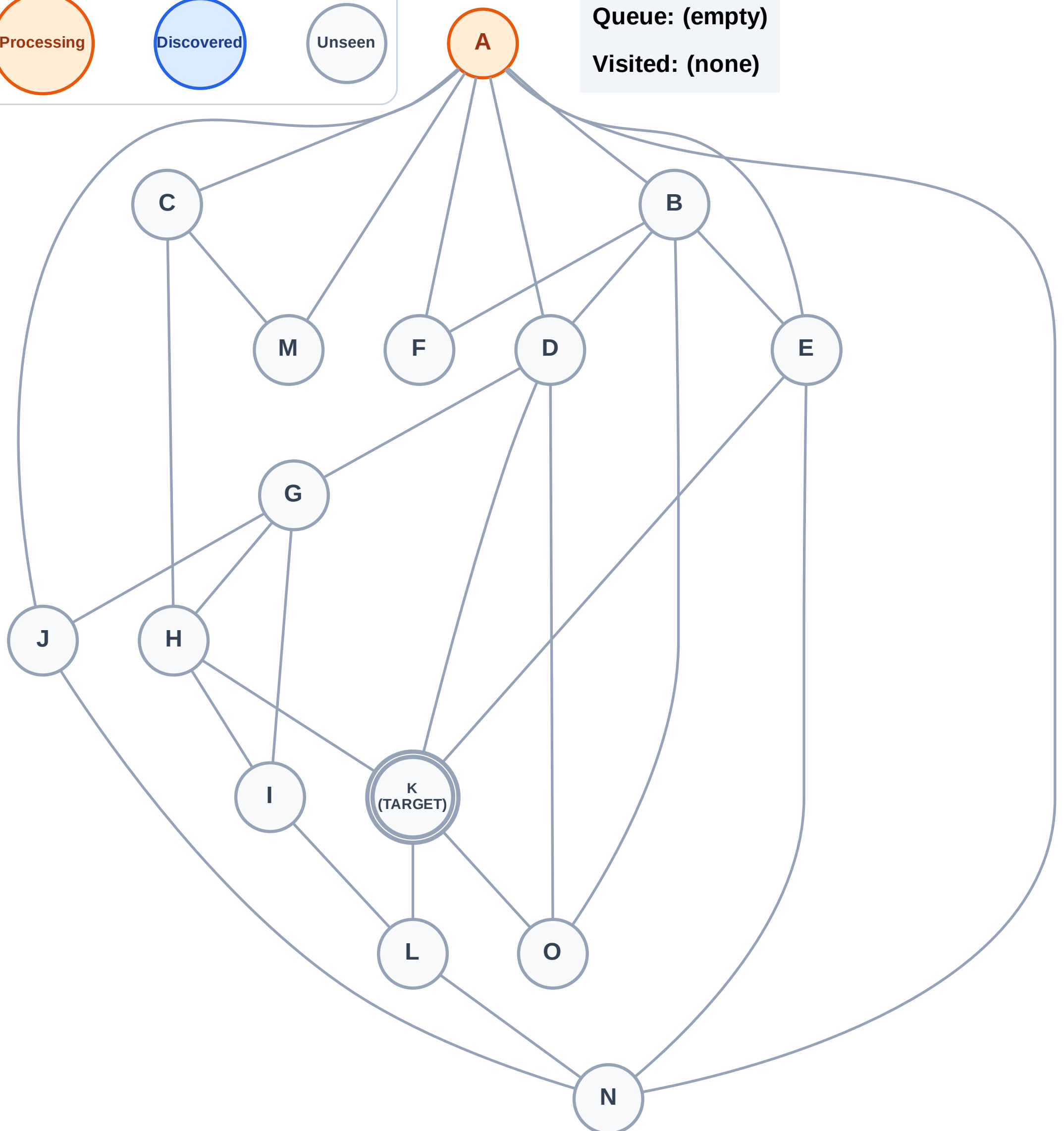
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

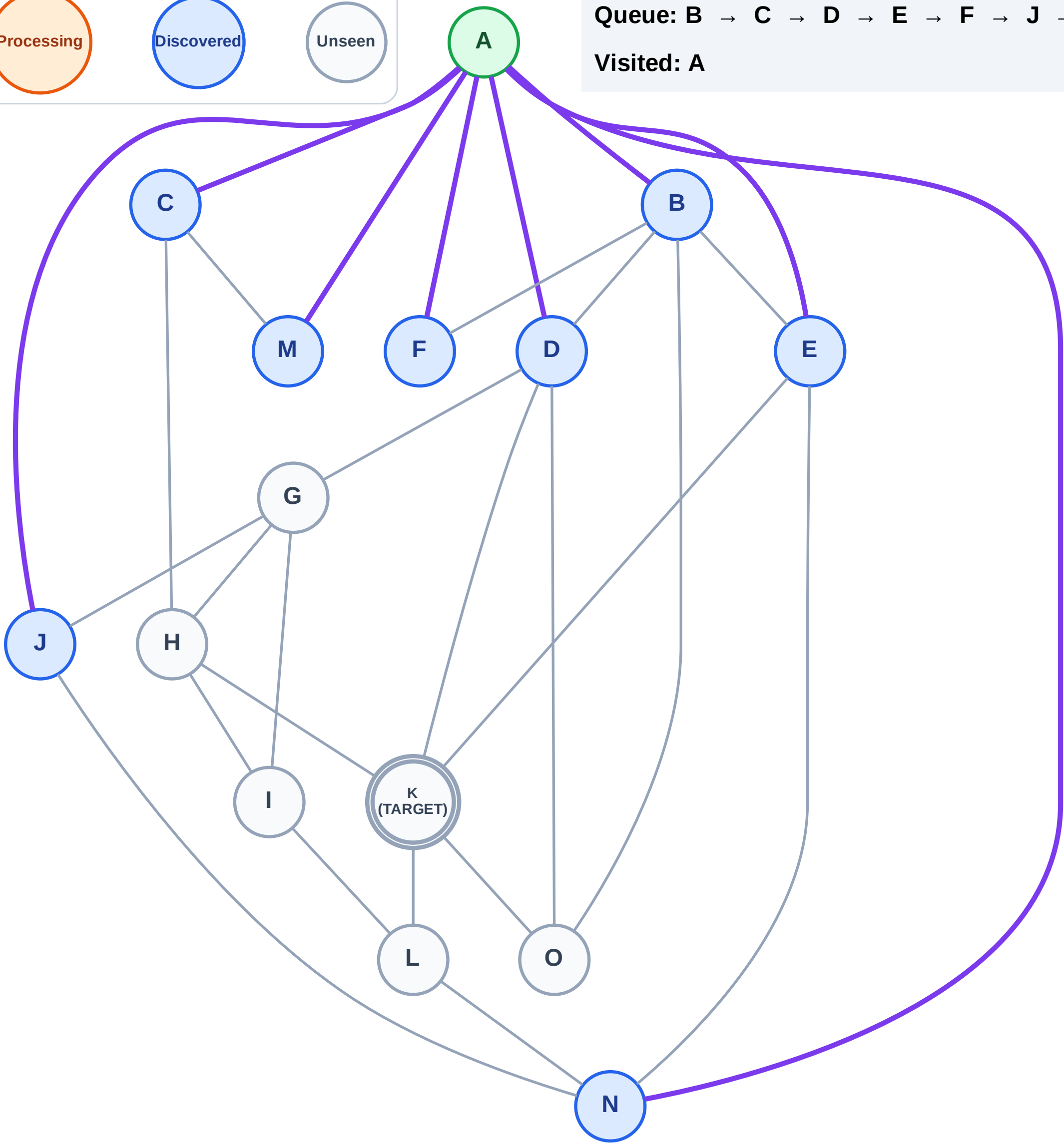
Step 3: Discover B, C, D, E, F, J, M, N from A

Legend



Queue: B → C → D → E → F → J → M → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

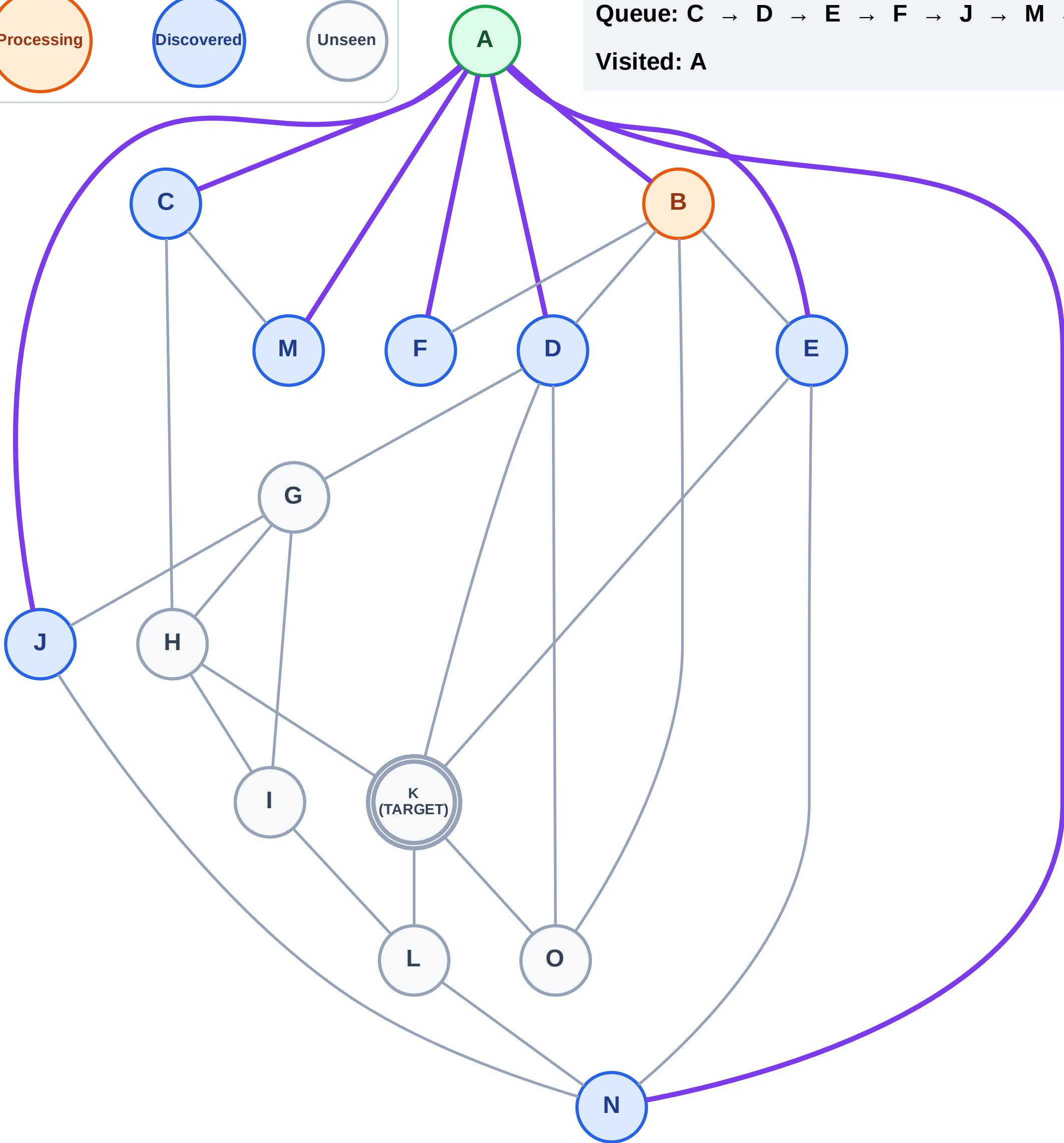
Processing

Discovered

Unseen

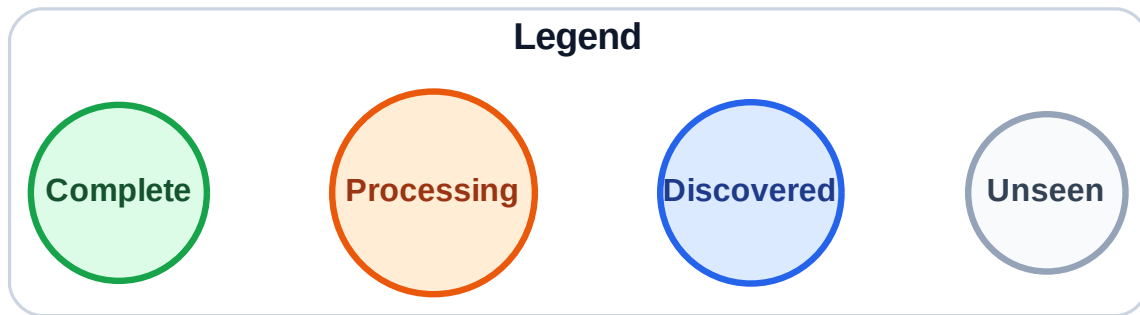
Queue: C → D → E → F → J → M → N

Visited: A



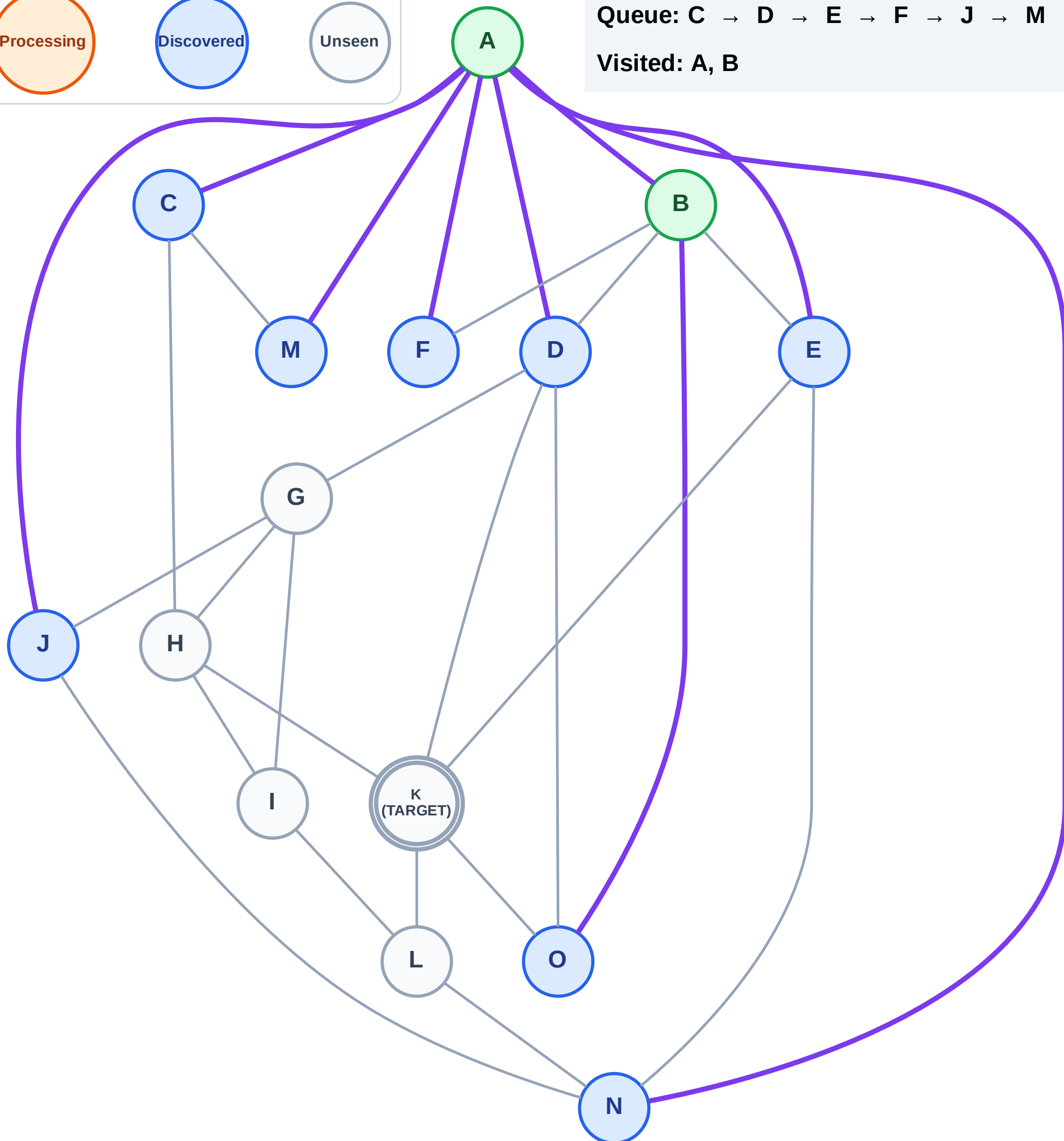
BFS Traversal

Step 5: Discover O from B



Queue: C → D → E → F → J → M → N → O

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

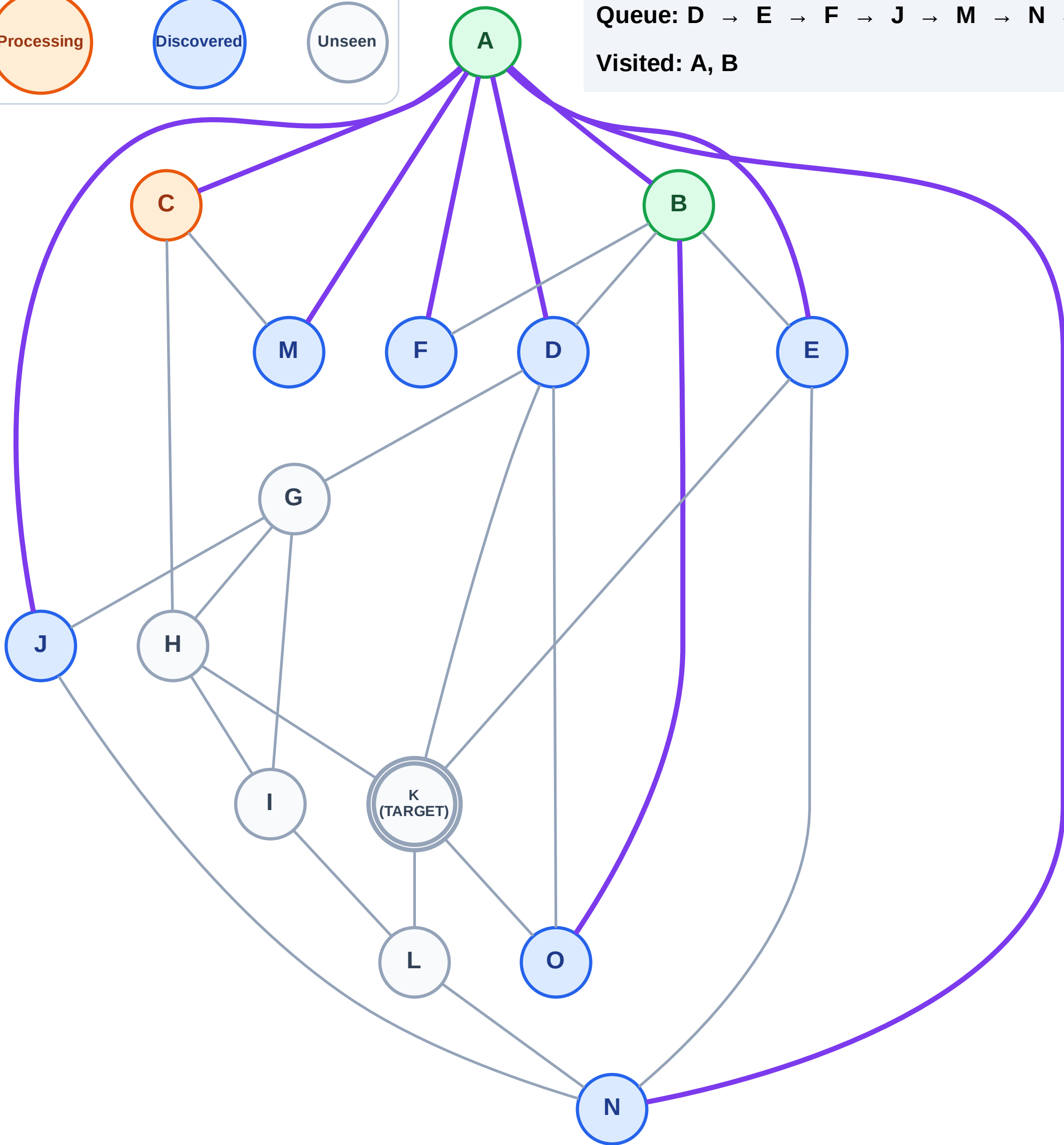
Processing

Discovered

Unseen

Queue: D → E → F → J → M → N → O

Visited: A, B



BFS Traversal

Step 7: Discover H from C

Legend

Complete

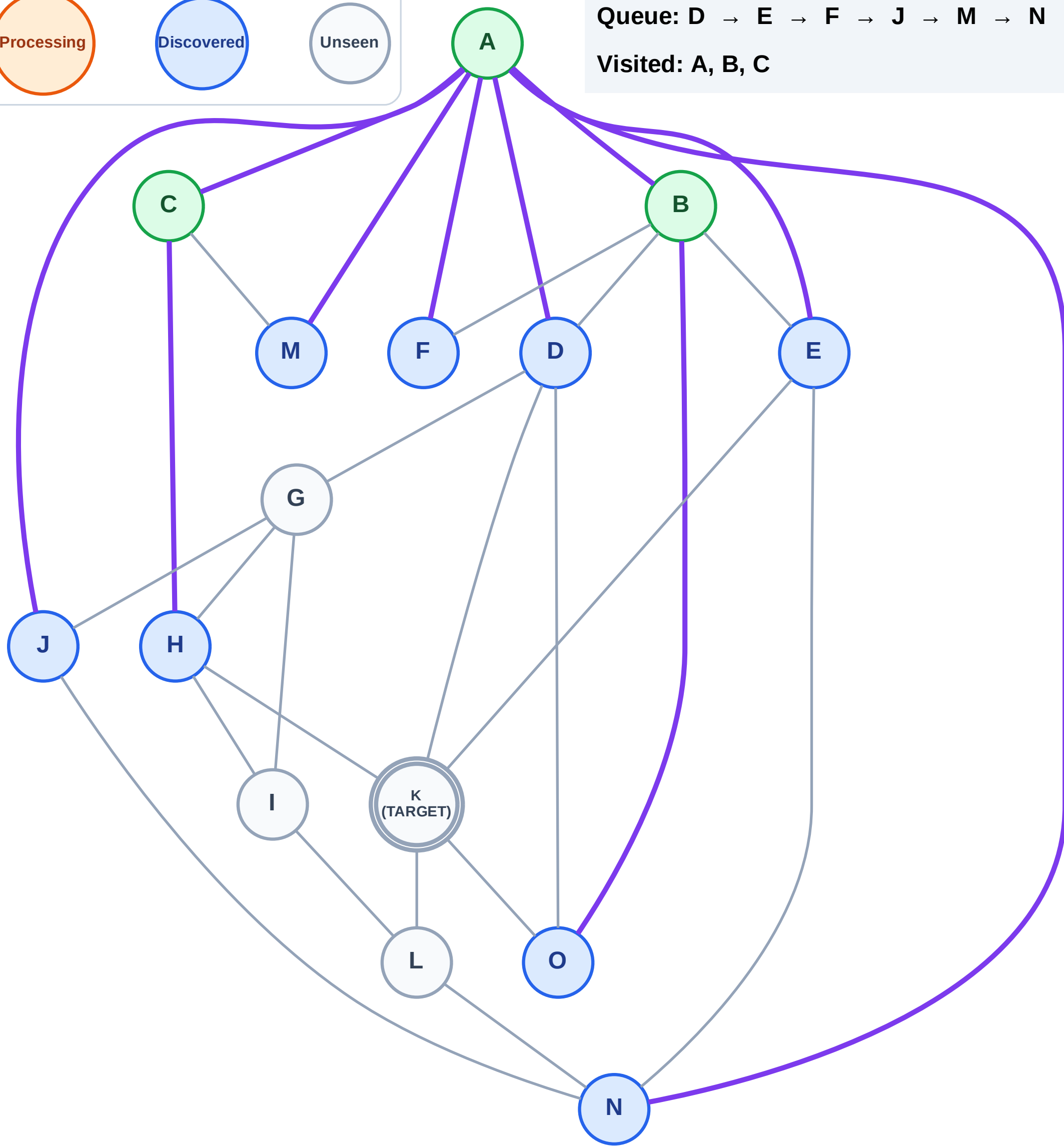
Processing

Discovered

Unseen

Queue: D → E → F → J → M → N → O → H

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

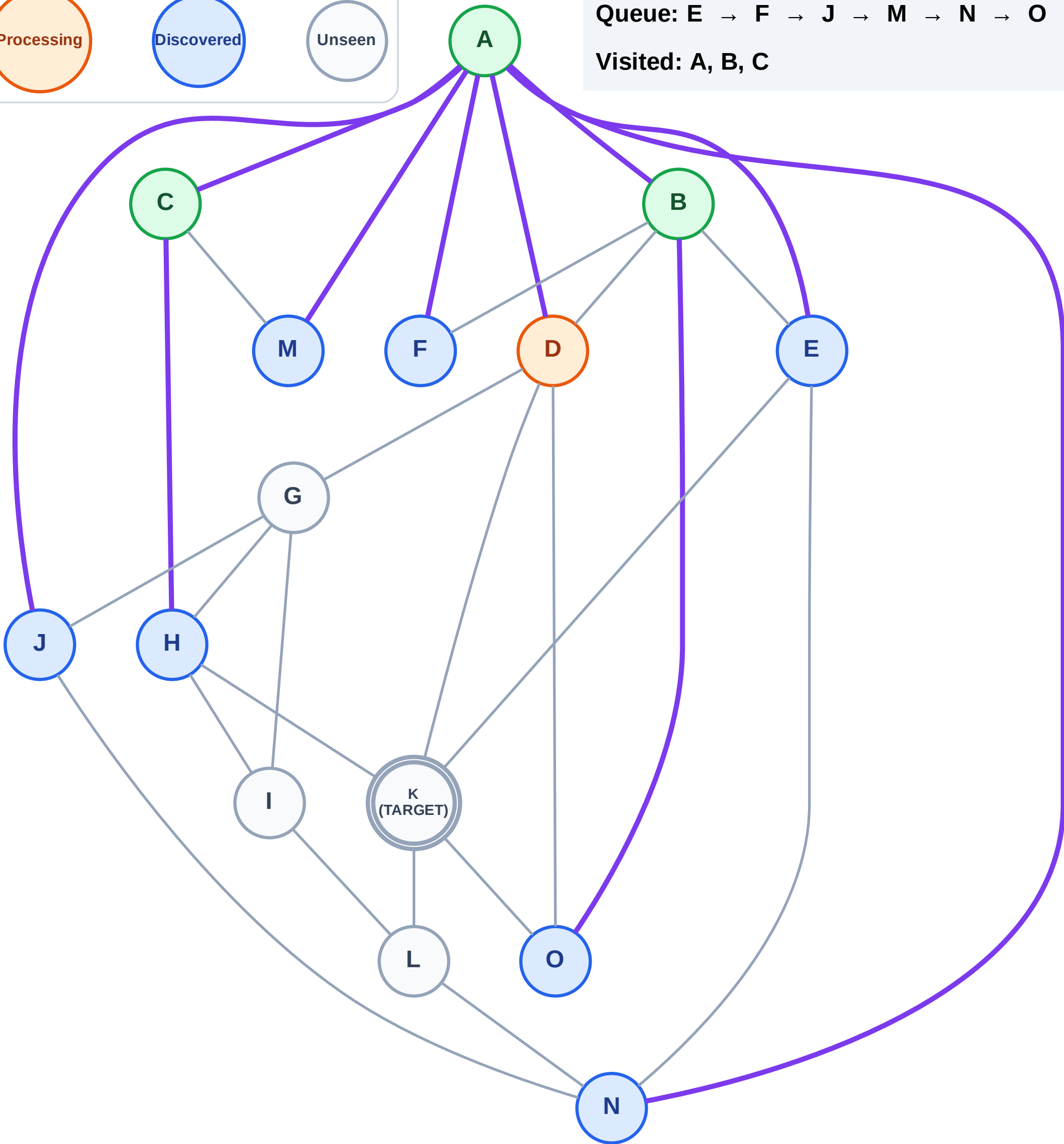
Processing

Discovered

Unseen

Queue: E → F → J → M → N → O → H

Visited: A, B, C



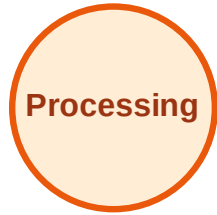
BFS Traversal

Step 9: Discover G, K from D

Legend



Complete



Processing



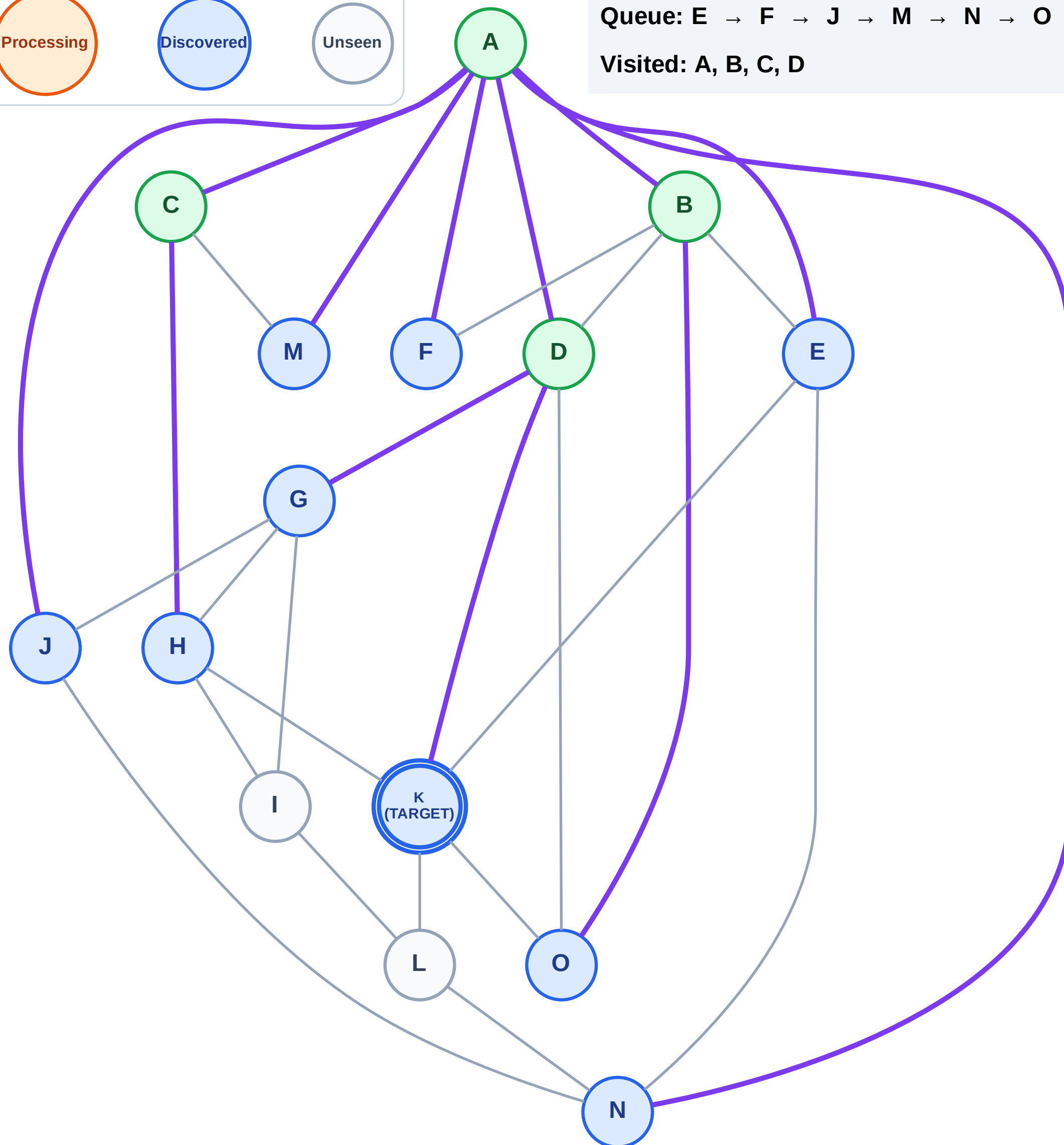
Discovered



Unseen

Queue: E → F → J → M → N → O → H → G → K

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

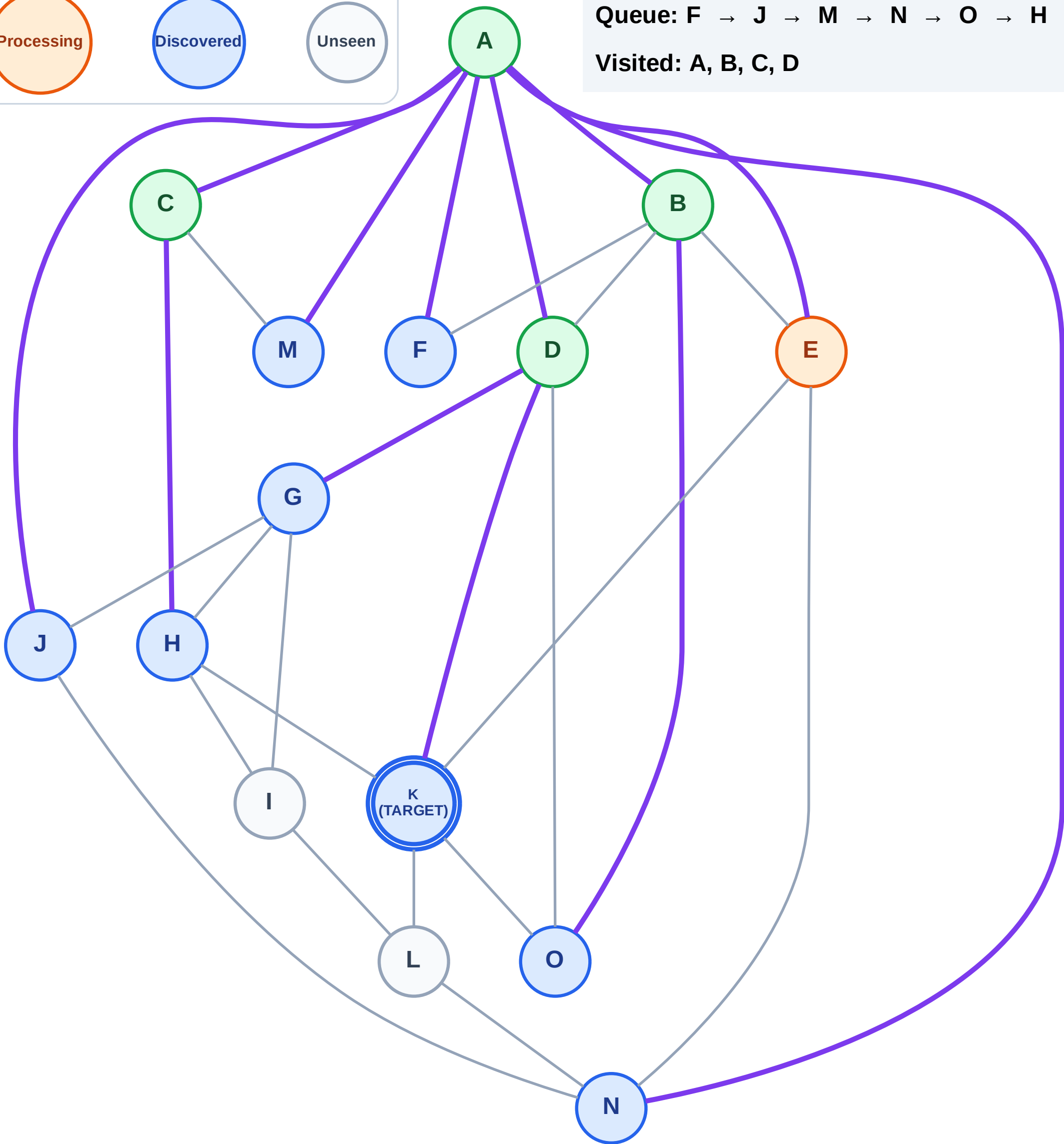
Processing

Discovered

Unseen

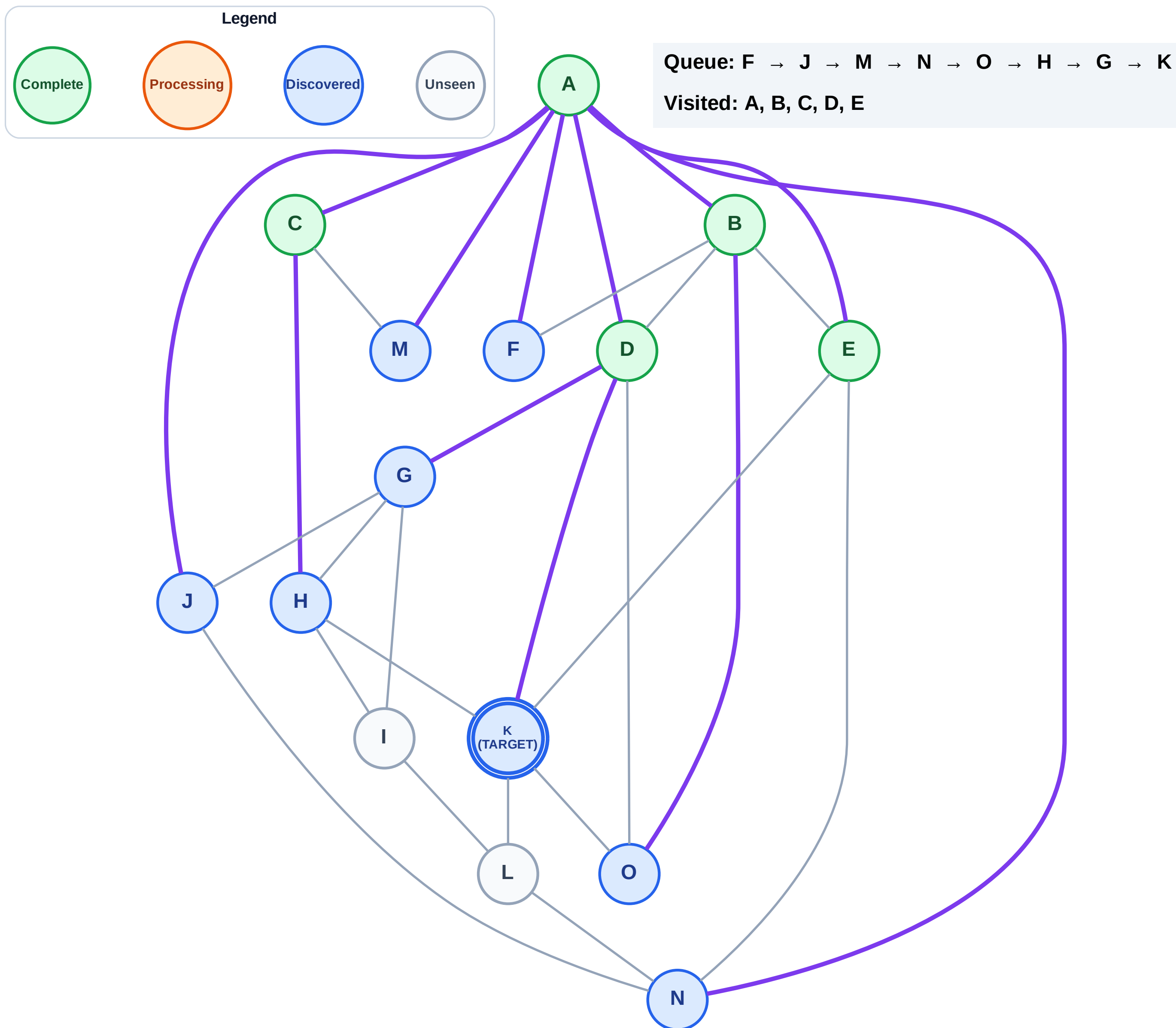
Queue: F → J → M → N → O → H → G → K

Visited: A, B, C, D



Step 11: No new neighbors from E

Step 11: No new neighbors from E



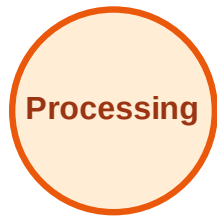
BFS Traversal

Step 12: Process node F

Legend



Complete



Processing



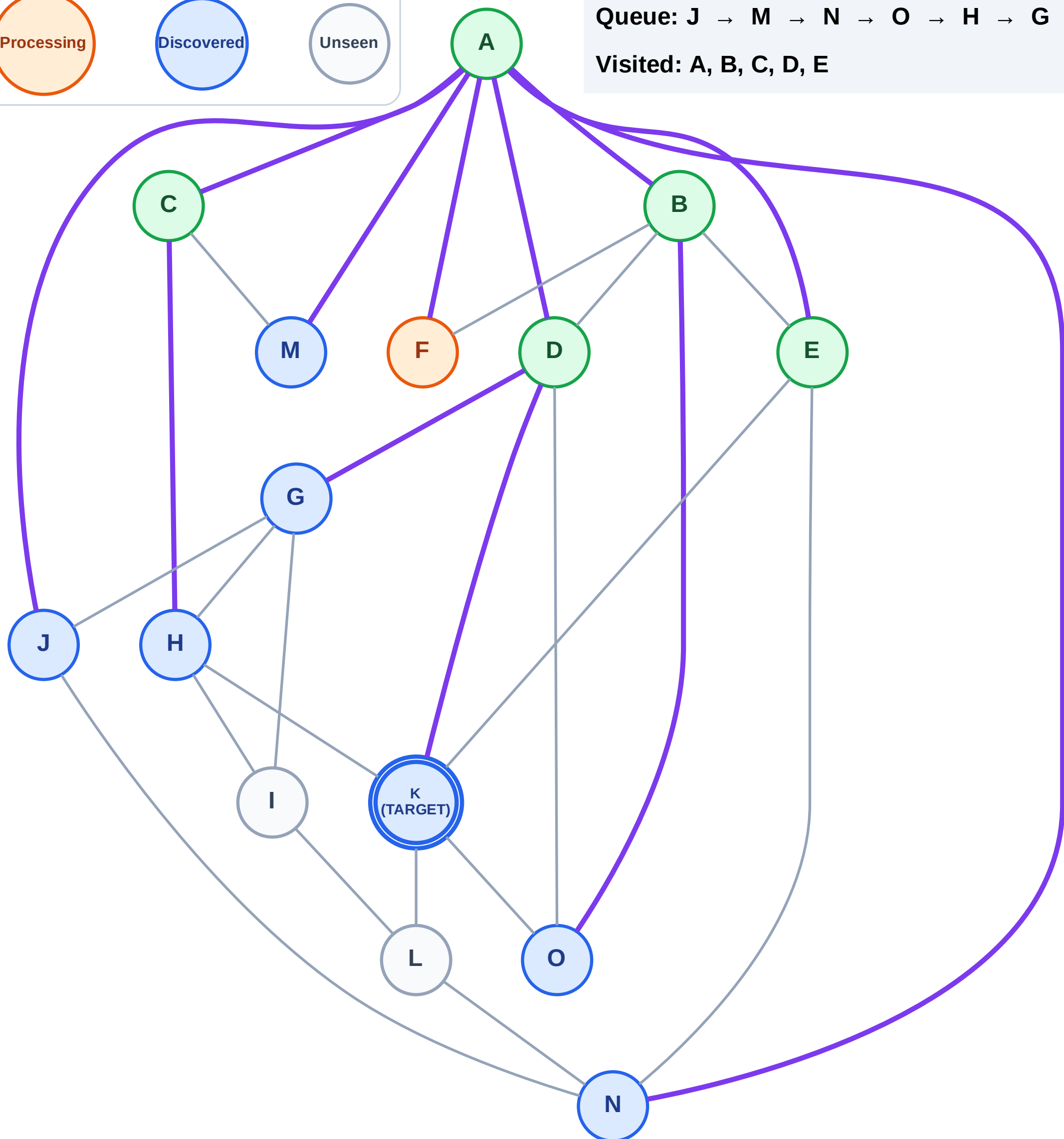
Discovered



Unseen

Queue: J → M → N → O → H → G → K

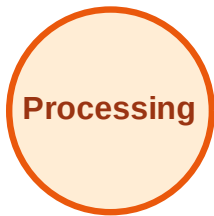
Visited: A, B, C, D, E



BFS Traversal

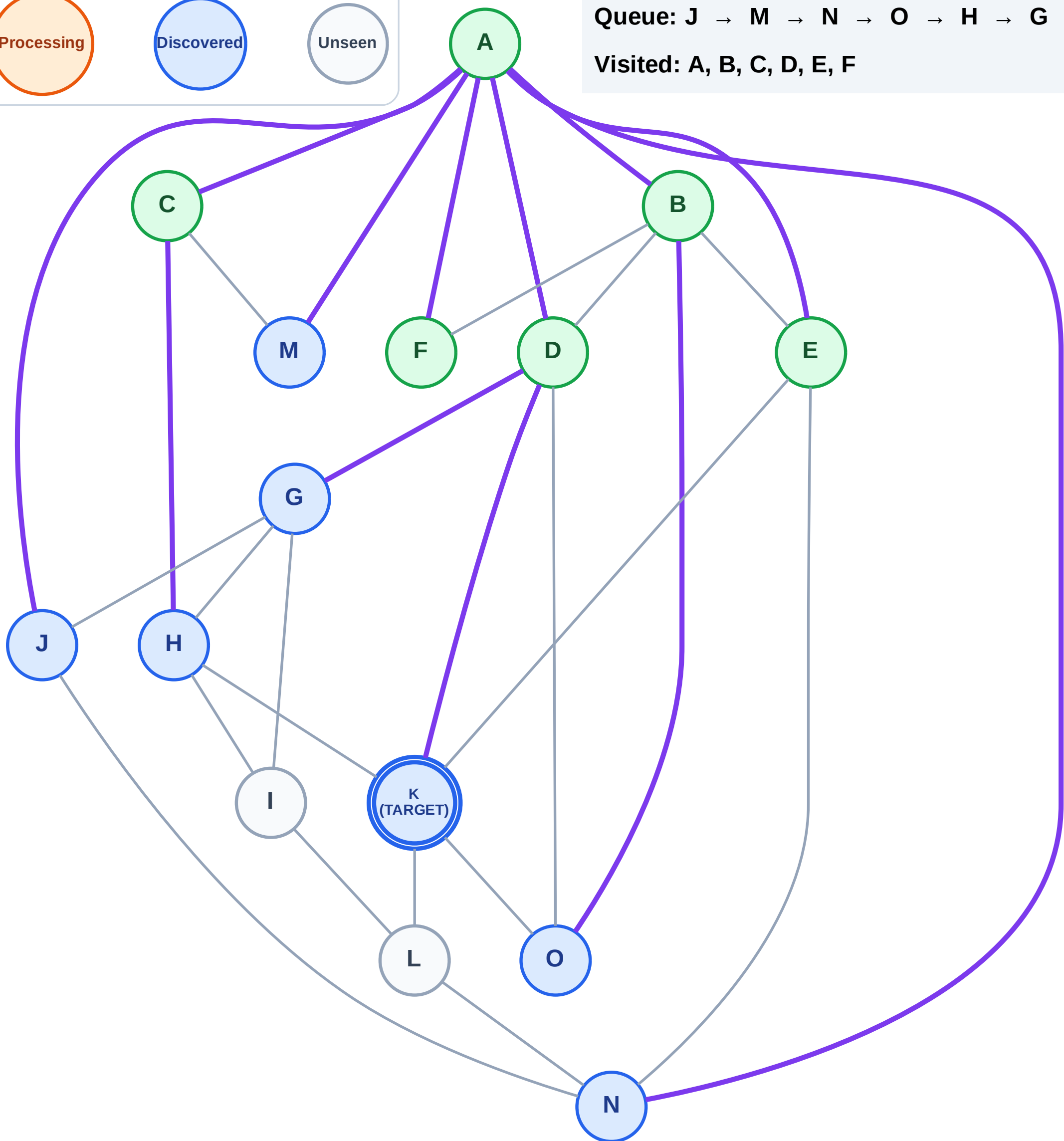
Step 13: No new neighbors from F

Legend



Queue: J → M → N → O → H → G → K

Visited: A, B, C, D, E, F



BFS Traversal

Step 14: Process node J

Legend

Complete

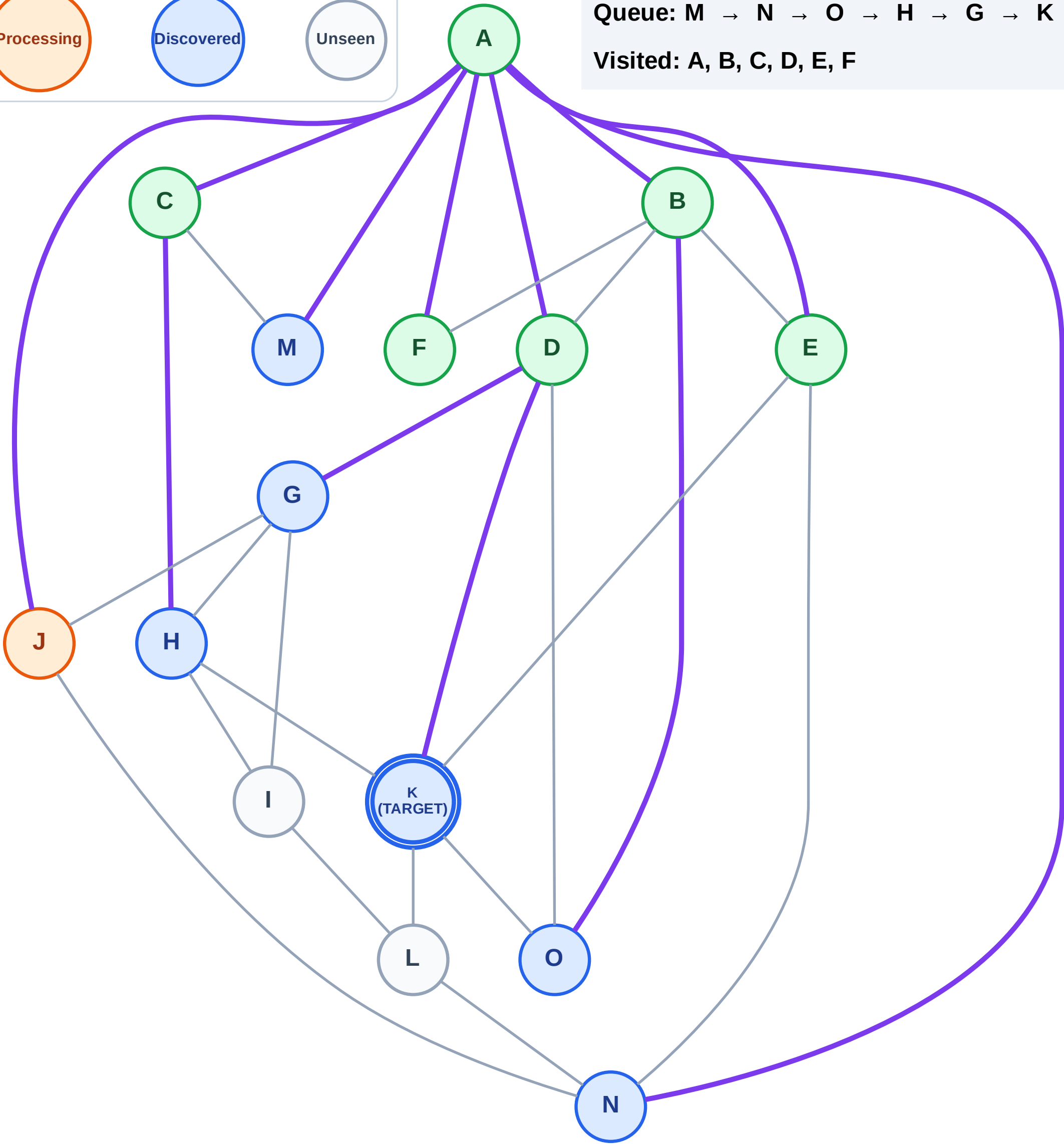
Processing

Discovered

Unseen

Queue: M → N → O → H → G → K

Visited: A, B, C, D, E, F



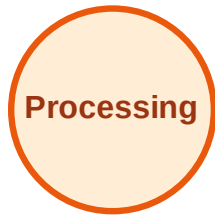
BFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing



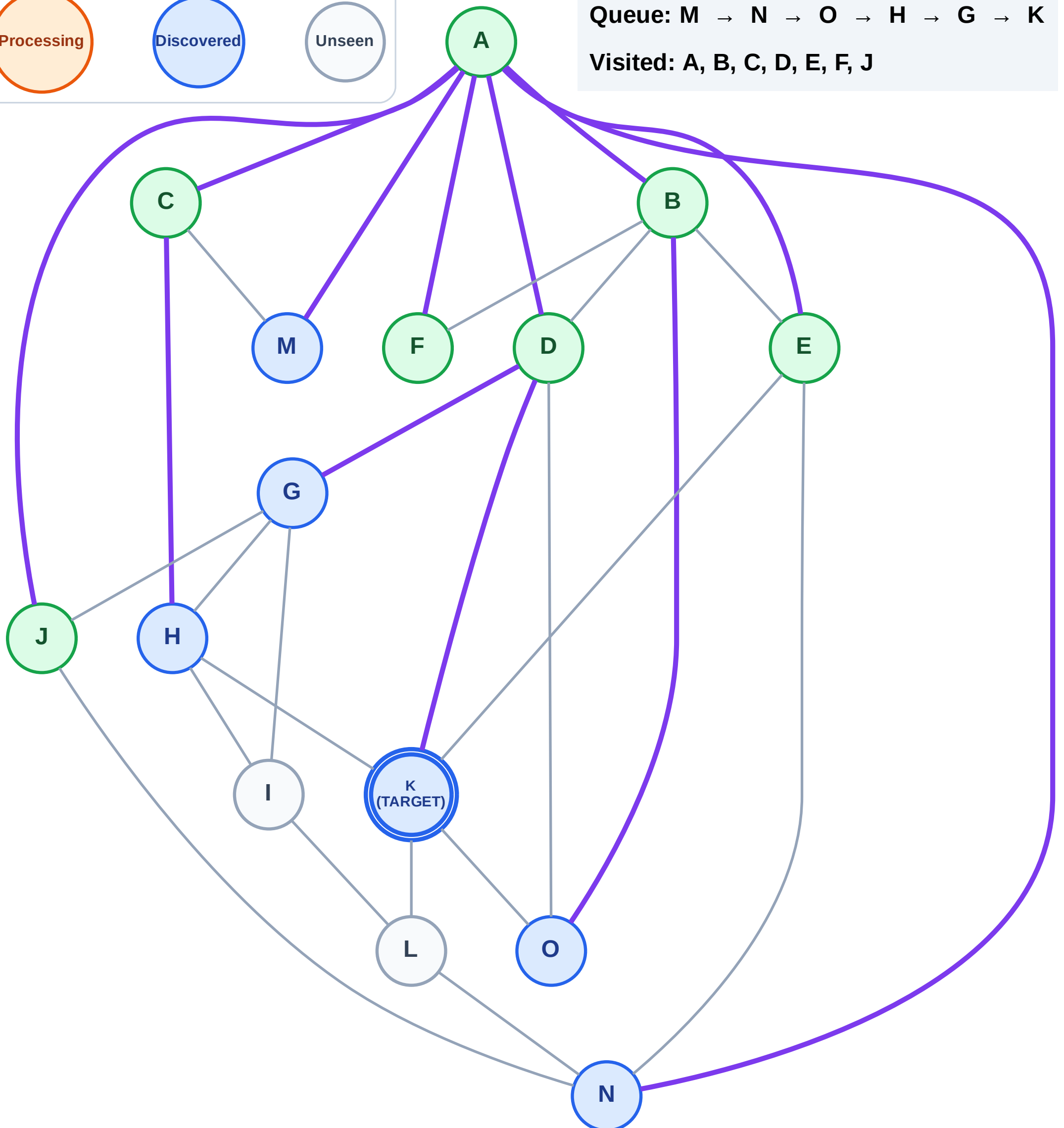
Discovered



Unseen

Queue: M → N → O → H → G → K

Visited: A, B, C, D, E, F, J



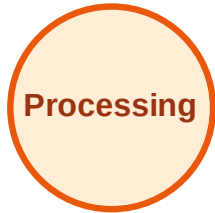
BFS Traversal

Step 16: Process node M

Legend



Complete



Processing



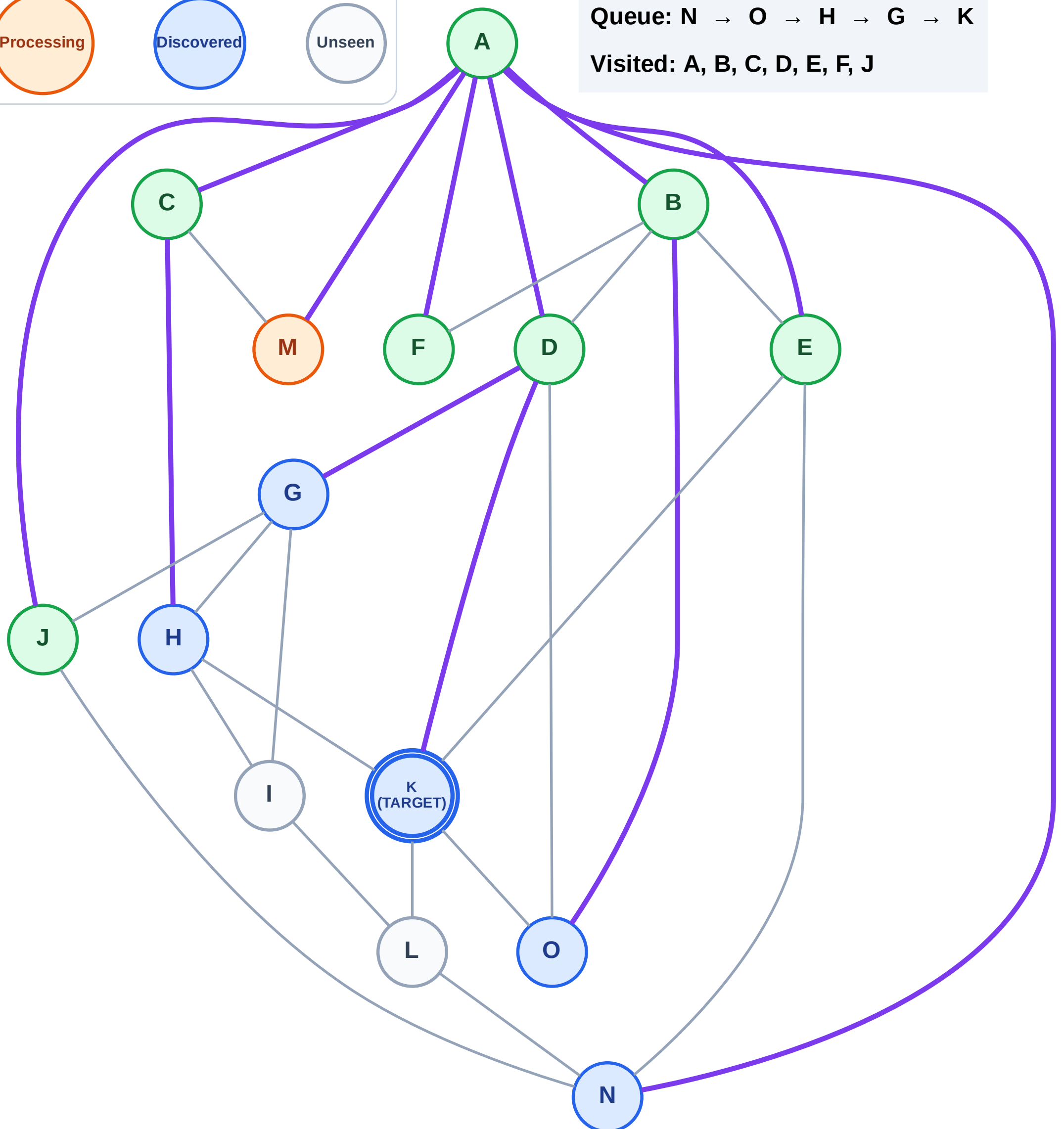
Discovered



Unseen

Queue: N → O → H → G → K

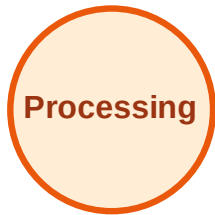
Visited: A, B, C, D, E, F, J



BFS Traversal

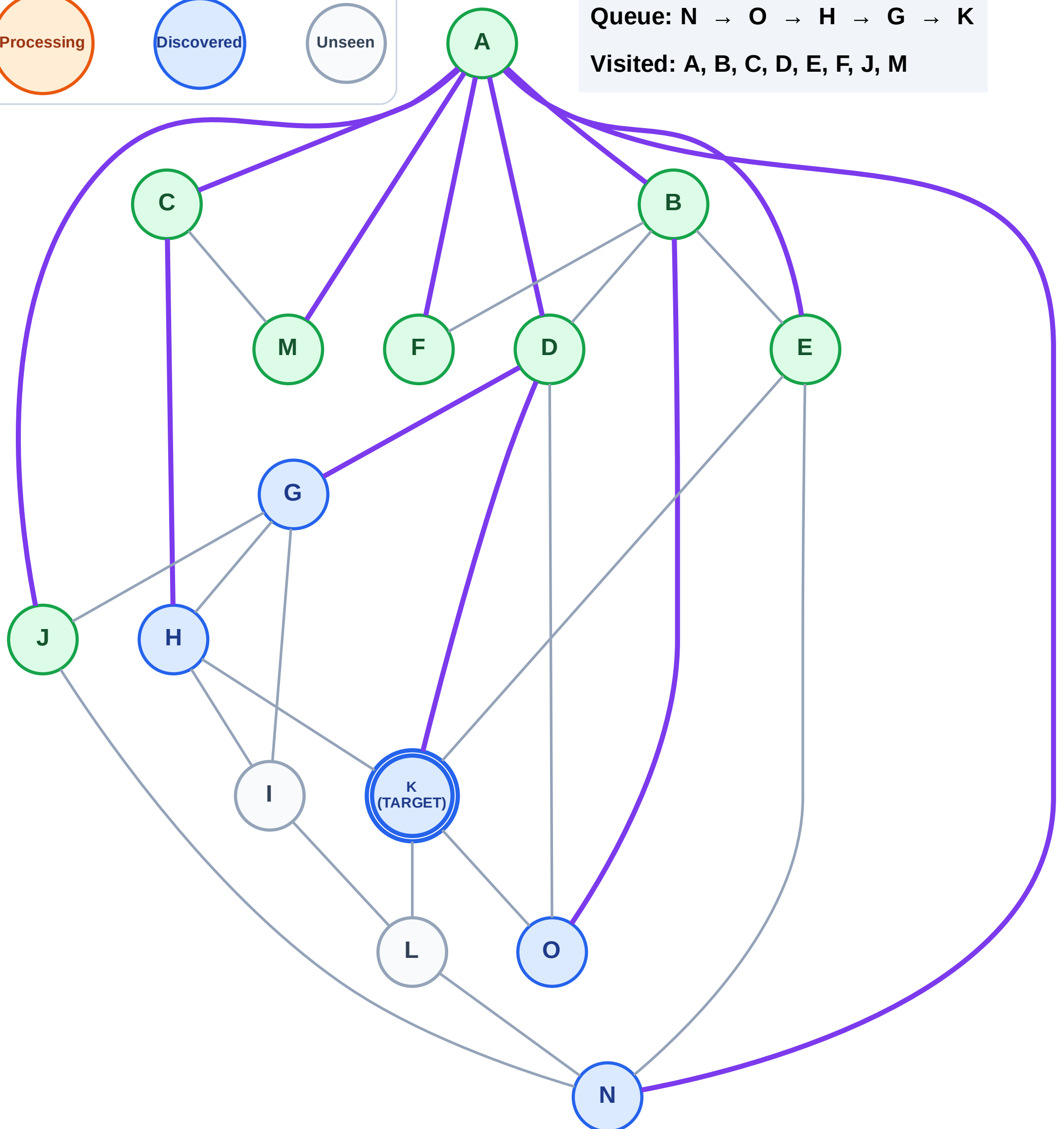
Step 17: No new neighbors from M

Legend



Queue: N → O → H → G → K

Visited: A, B, C, D, E, F, J, M



BFS Traversal

Step 18: Process node N

Legend

Complete

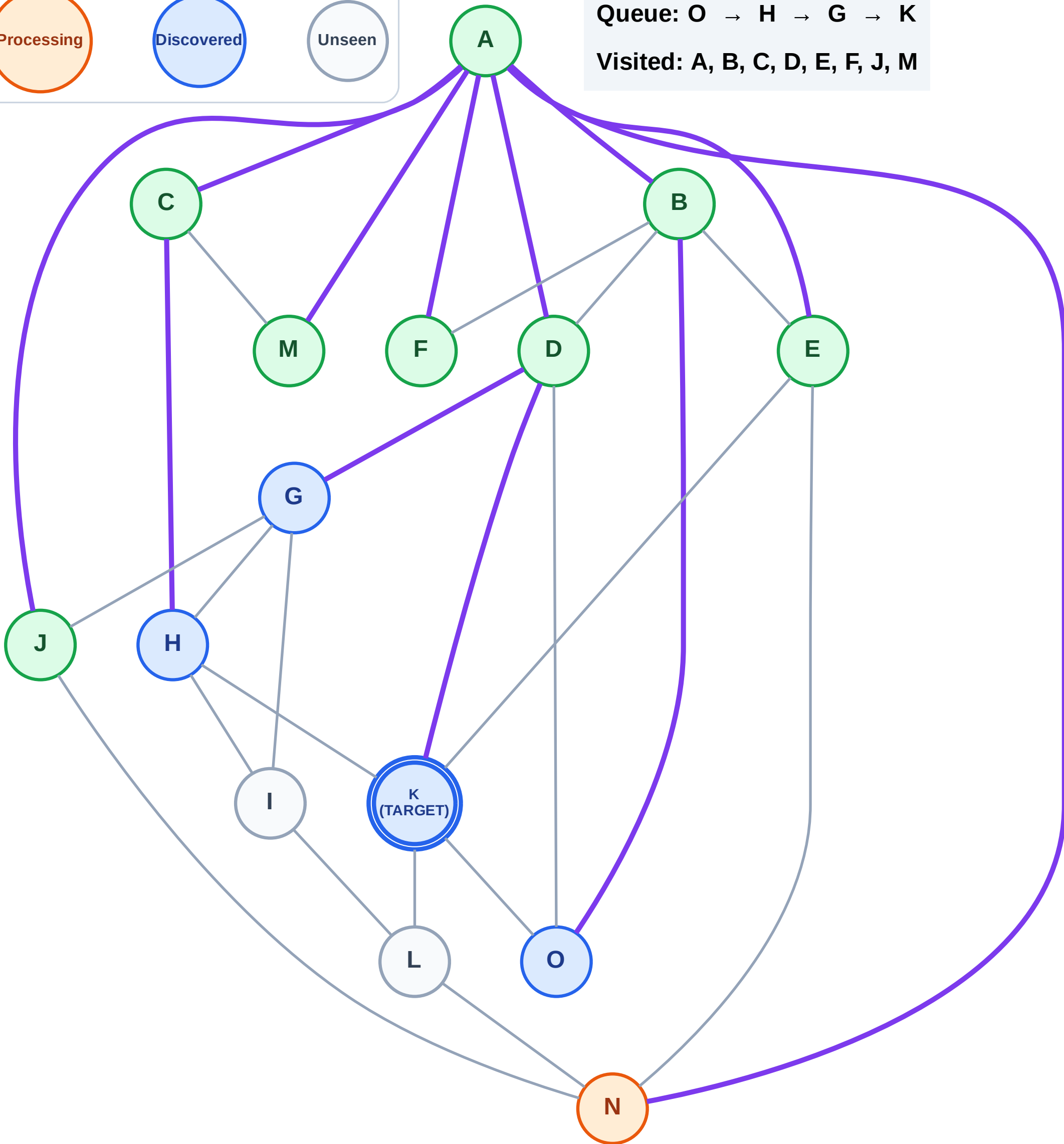
Processing

Discovered

Unseen

Queue: O → H → G → K

Visited: A, B, C, D, E, F, J, M



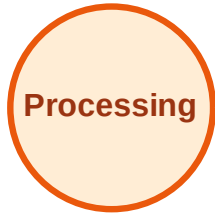
BFS Traversal

Step 19: Discover L from N

Legend



Complete



Processing



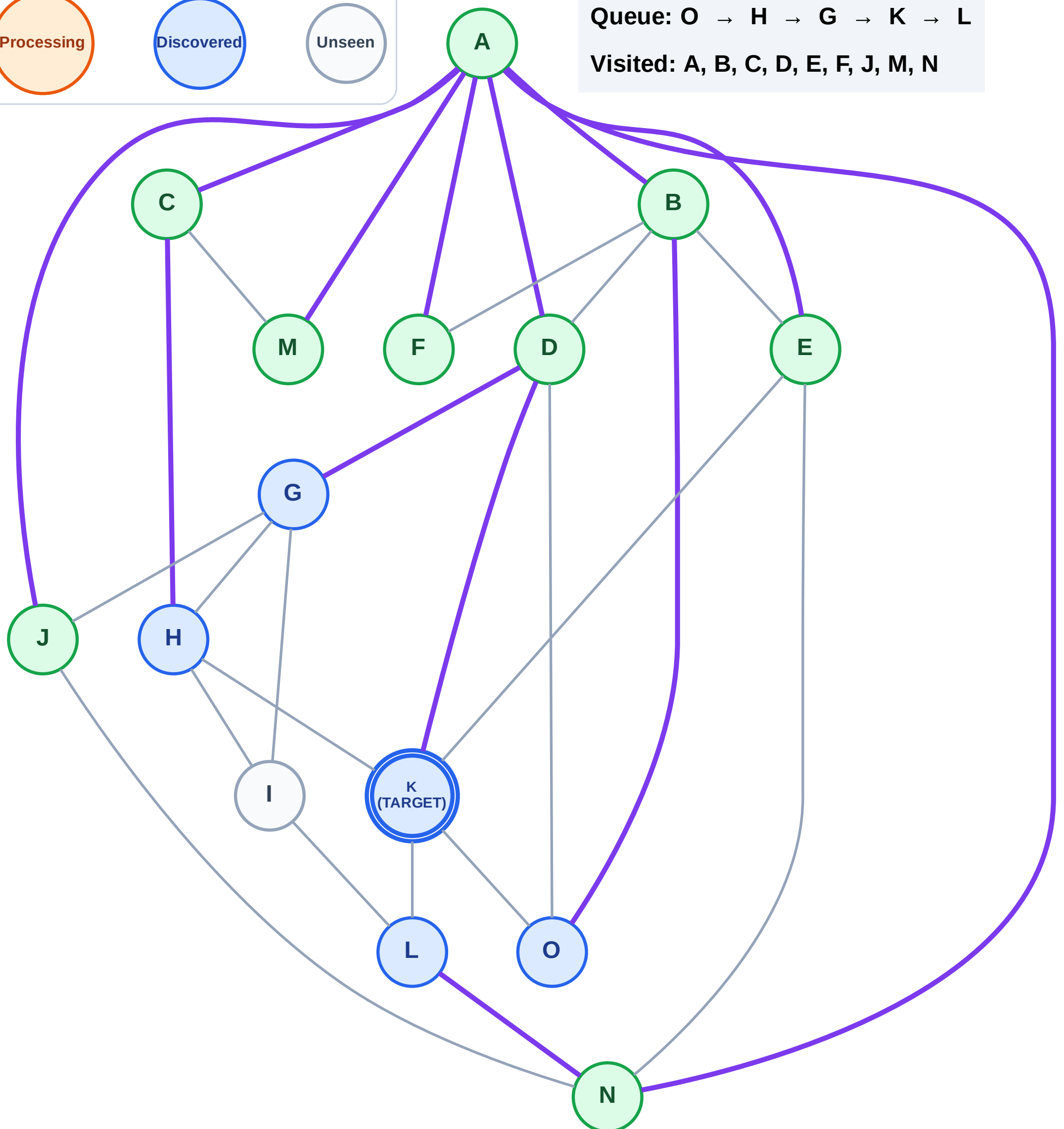
Discovered



Unseen

Queue: O → H → G → K → L

Visited: A, B, C, D, E, F, J, M, N



BFS Traversal

Step 20: Process node O

Legend

Complete

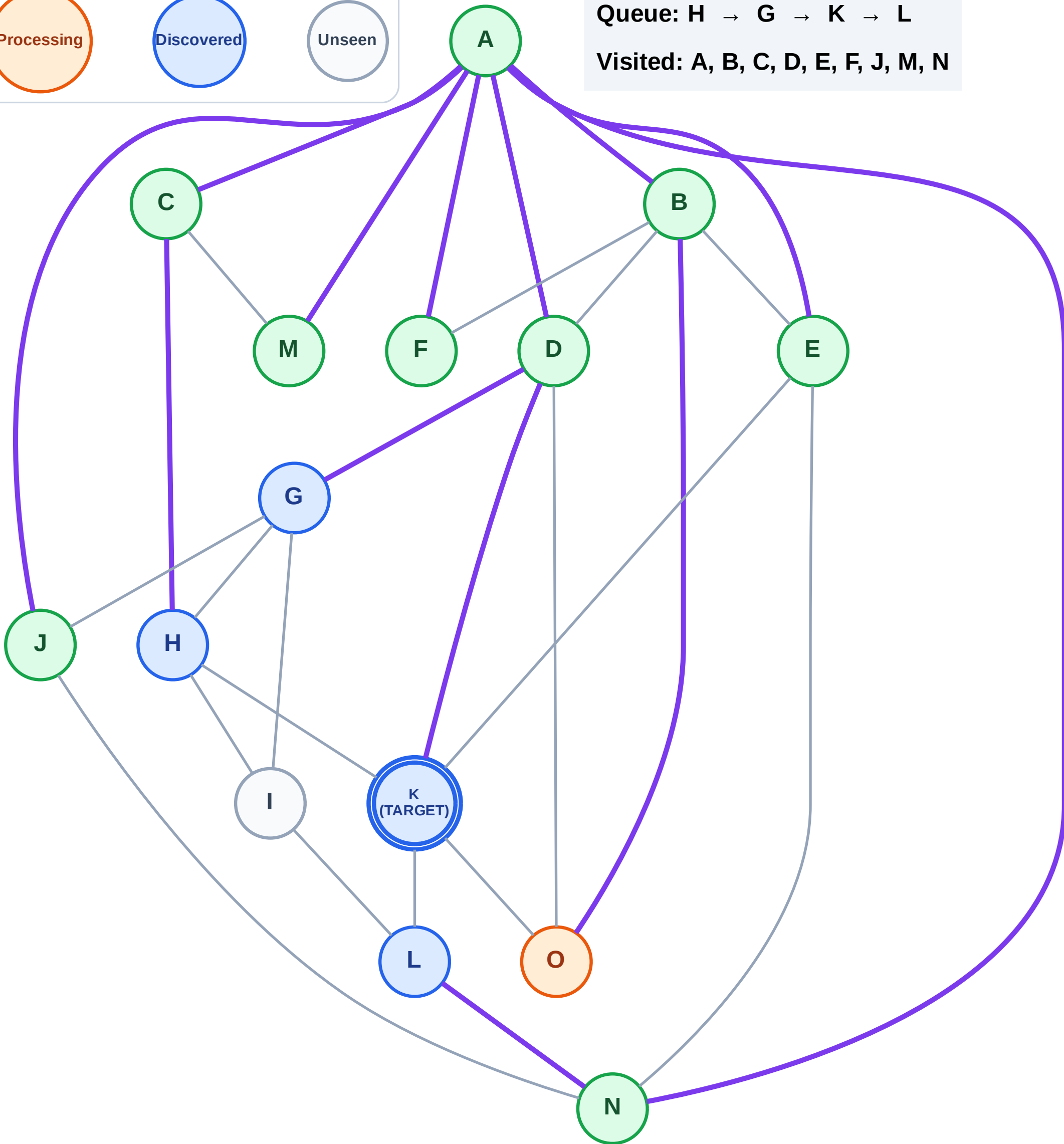
Processing

Discovered

Unseen

Queue: H → G → K → L

Visited: A, B, C, D, E, F, J, M, N



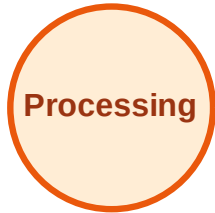
BFS Traversal

Step 21: No new neighbors from O

Legend



Complete



Processing



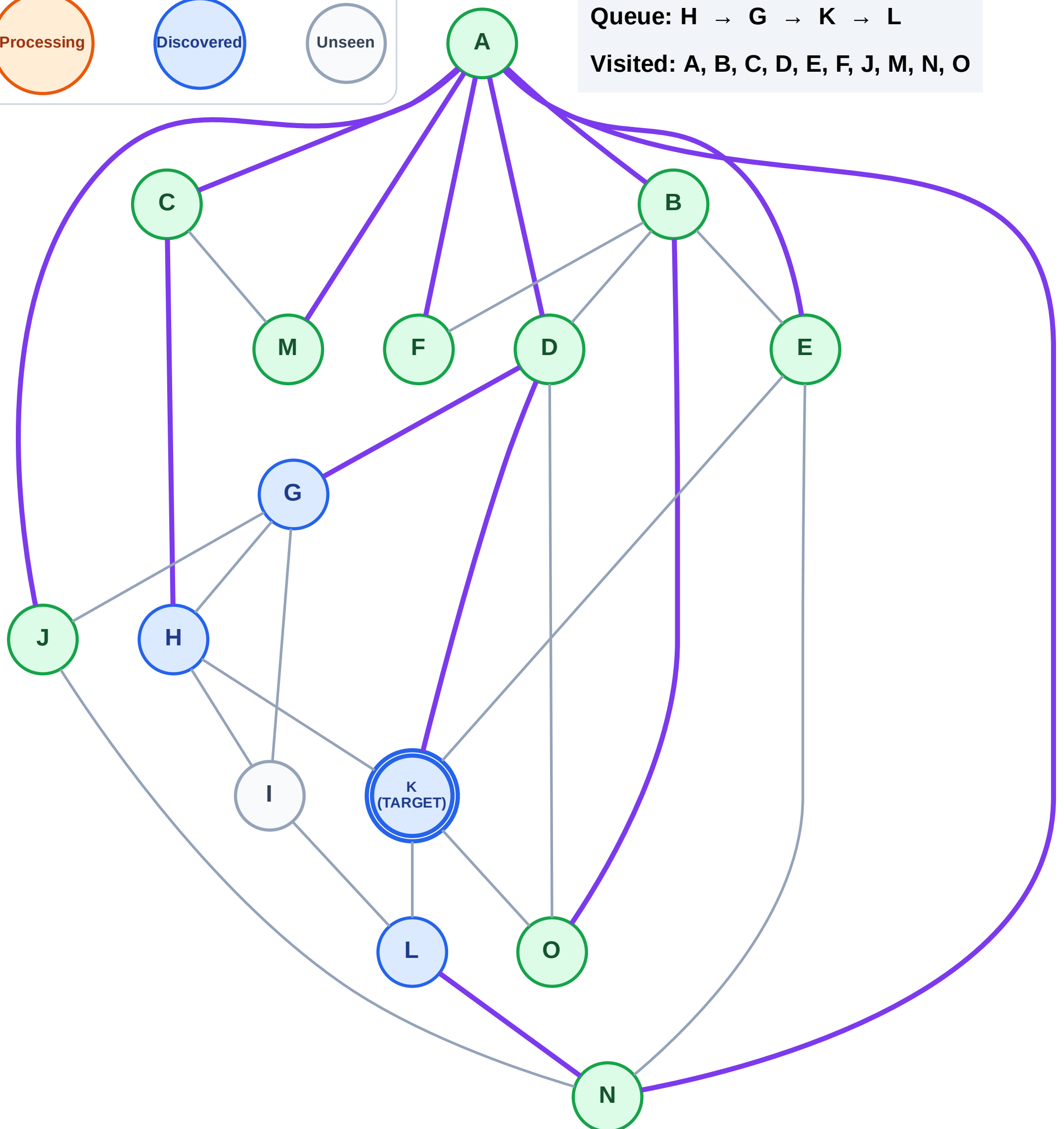
Discovered



Unseen

Queue: H → G → K → L

Visited: A, B, C, D, E, F, J, M, N, O



BFS Traversal

Step 22: Process node H

Legend

Complete

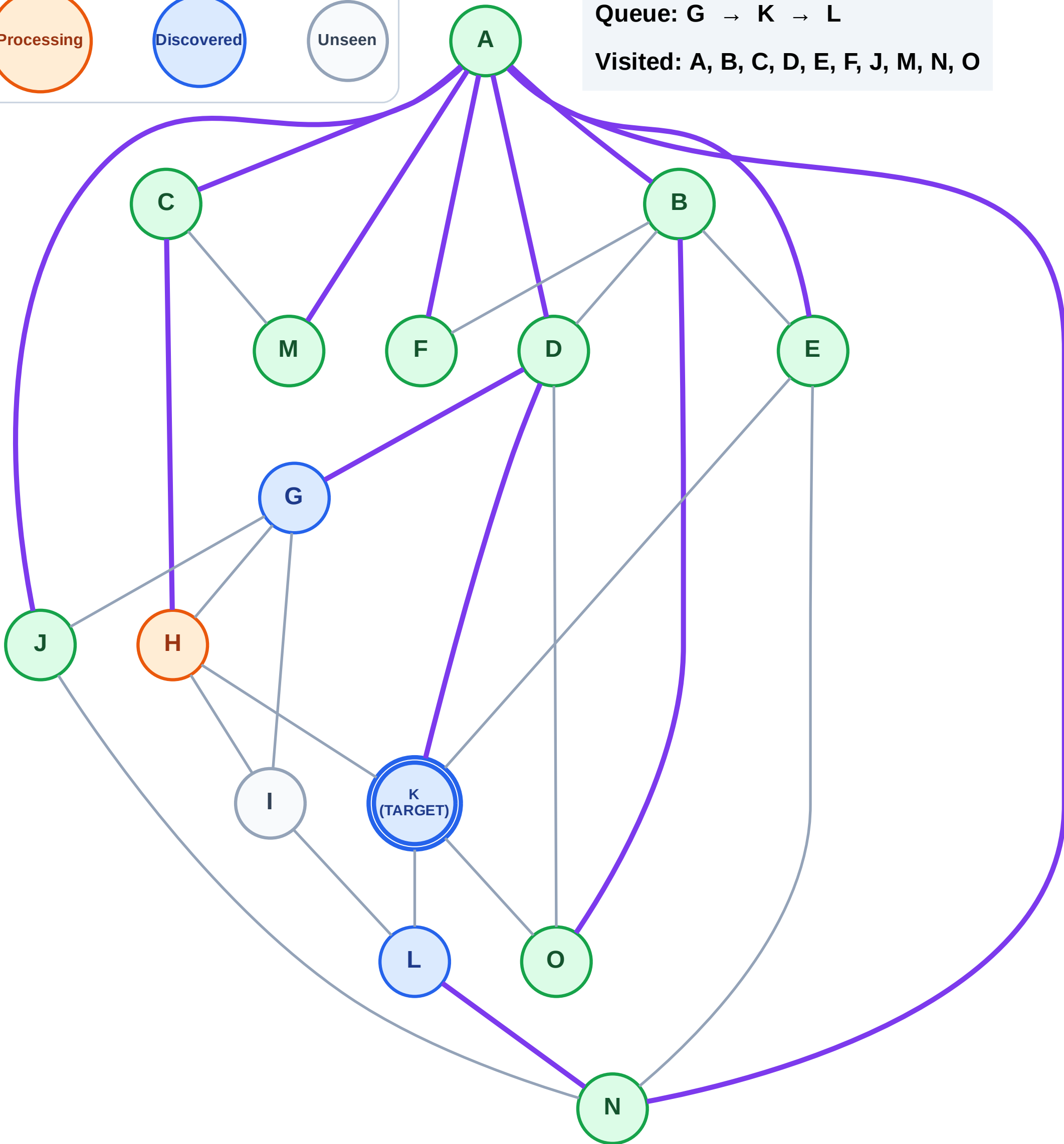
Processing

Discovered

Unseen

Queue: G → K → L

Visited: A, B, C, D, E, F, J, M, N, O



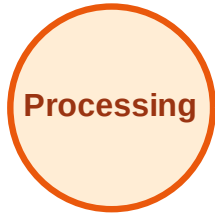
BFS Traversal

Step 23: Discover I from H

Legend



Complete



Processing



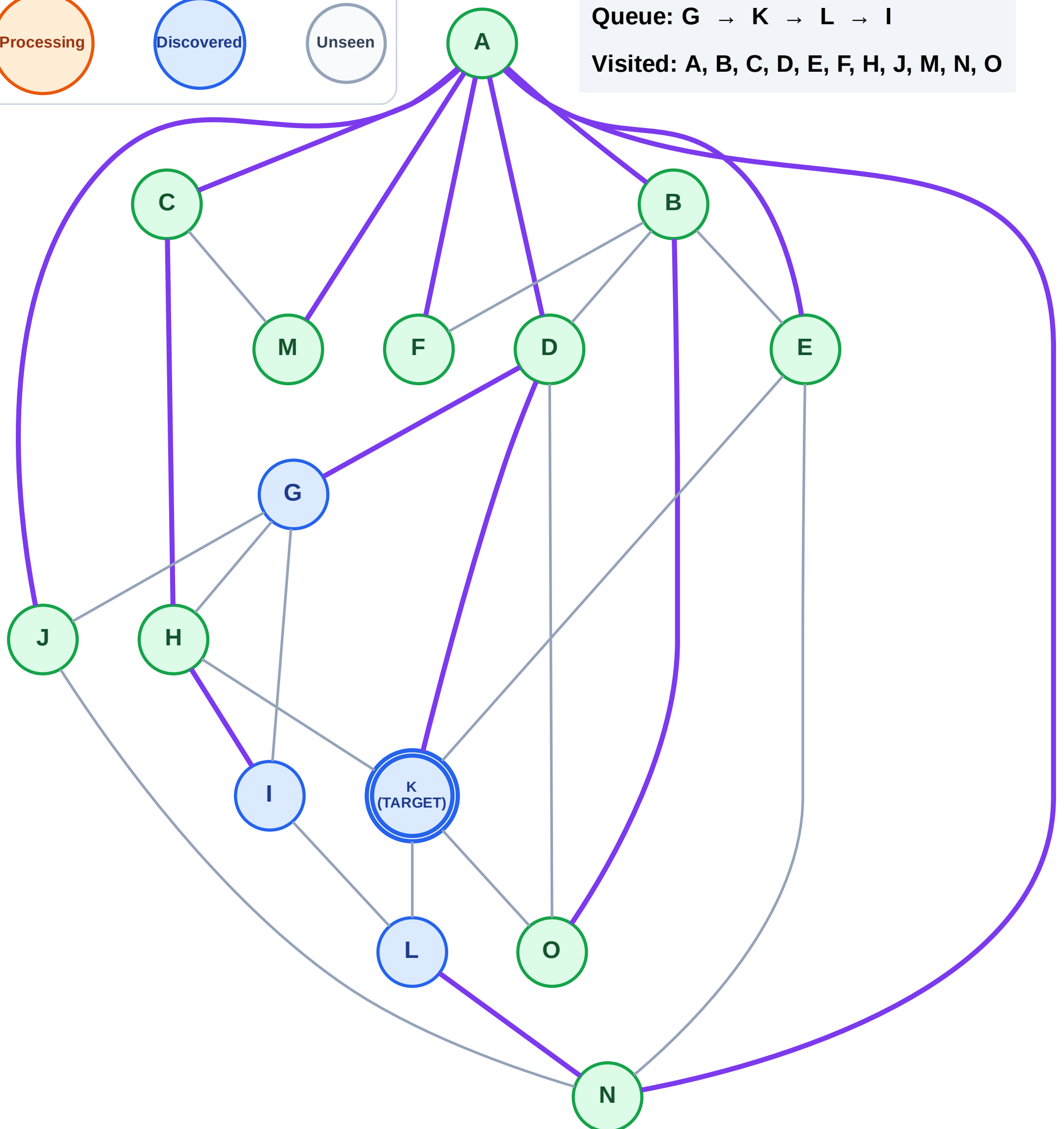
Discovered



Unseen

Queue: G → K → L → I

Visited: A, B, C, D, E, F, H, J, M, N, O



BFS Traversal

Step 24: Process node G

Legend

Complete

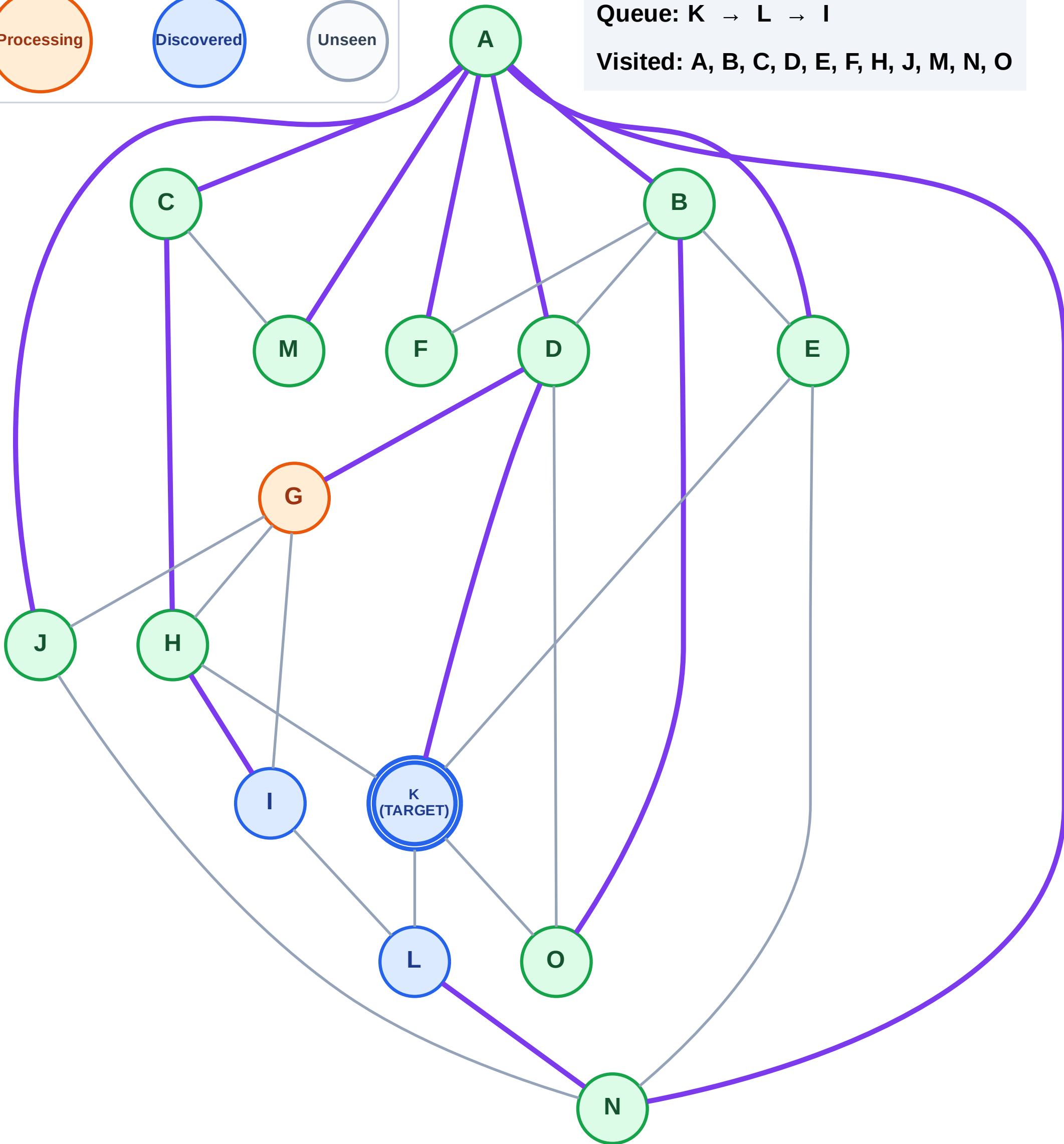
Processing

Discovered

Unseen

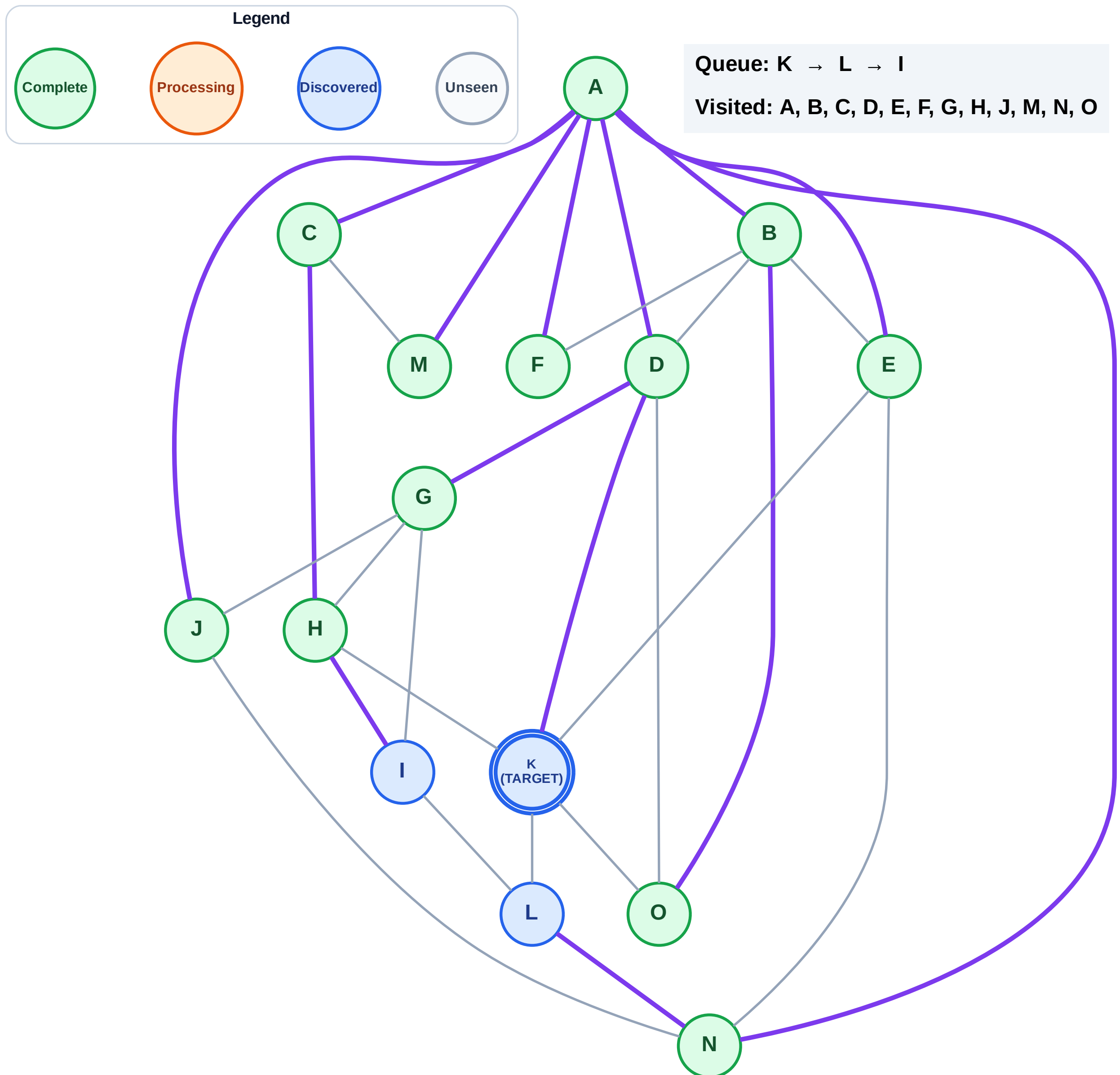
Queue: K → L → I

Visited: A, B, C, D, E, F, H, J, M, N, O



Step 25: No new neighbors from G

Step 25: No new neighbors from G



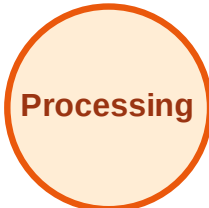
BFS Traversal

Step 26: Process node K

Legend



Complete



Processing



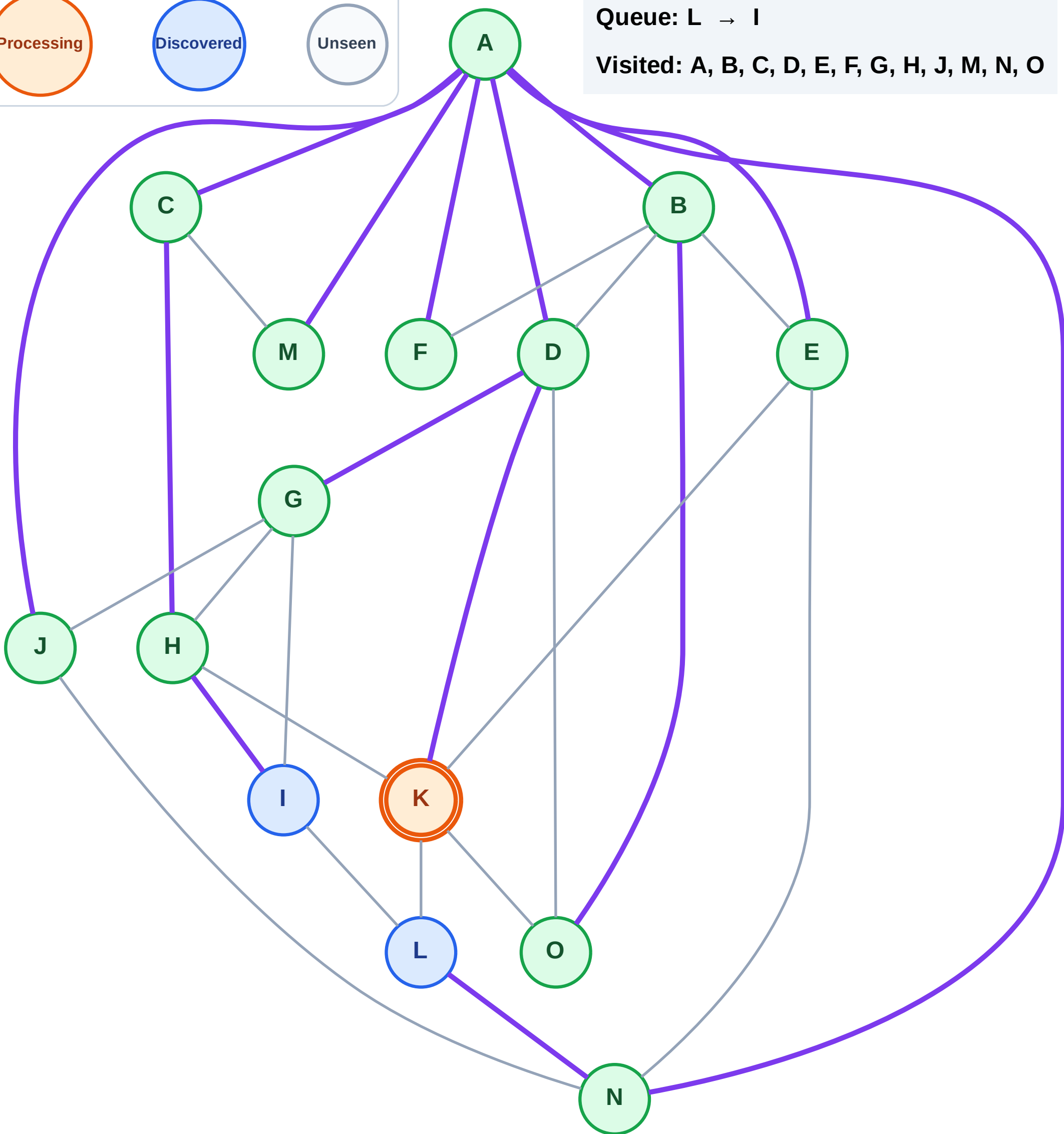
Discovered



Unseen

Queue: L → I

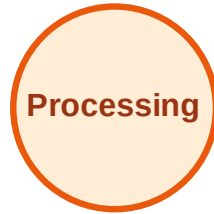
Visited: A, B, C, D, E, F, G, H, J, M, N, O



BFS Traversal

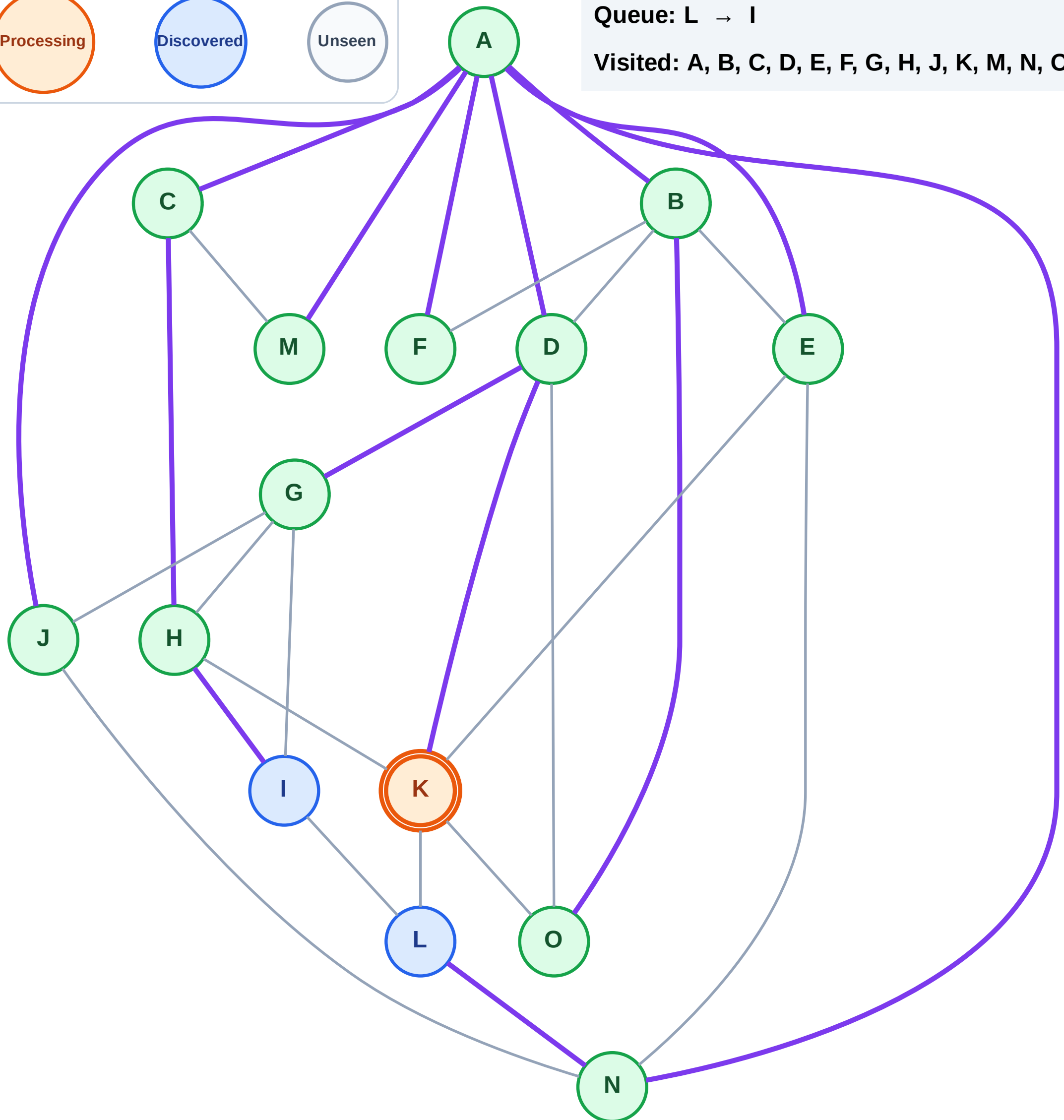
Step 27: TARGET NODE K REACHED! Path (2 hops): A → D → K

Legend



Queue: L → I

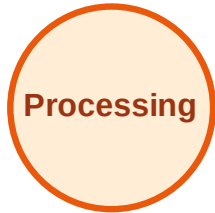
Visited: A, B, C, D, E, F, G, H, J, K, M, N, O



DFS Traversal

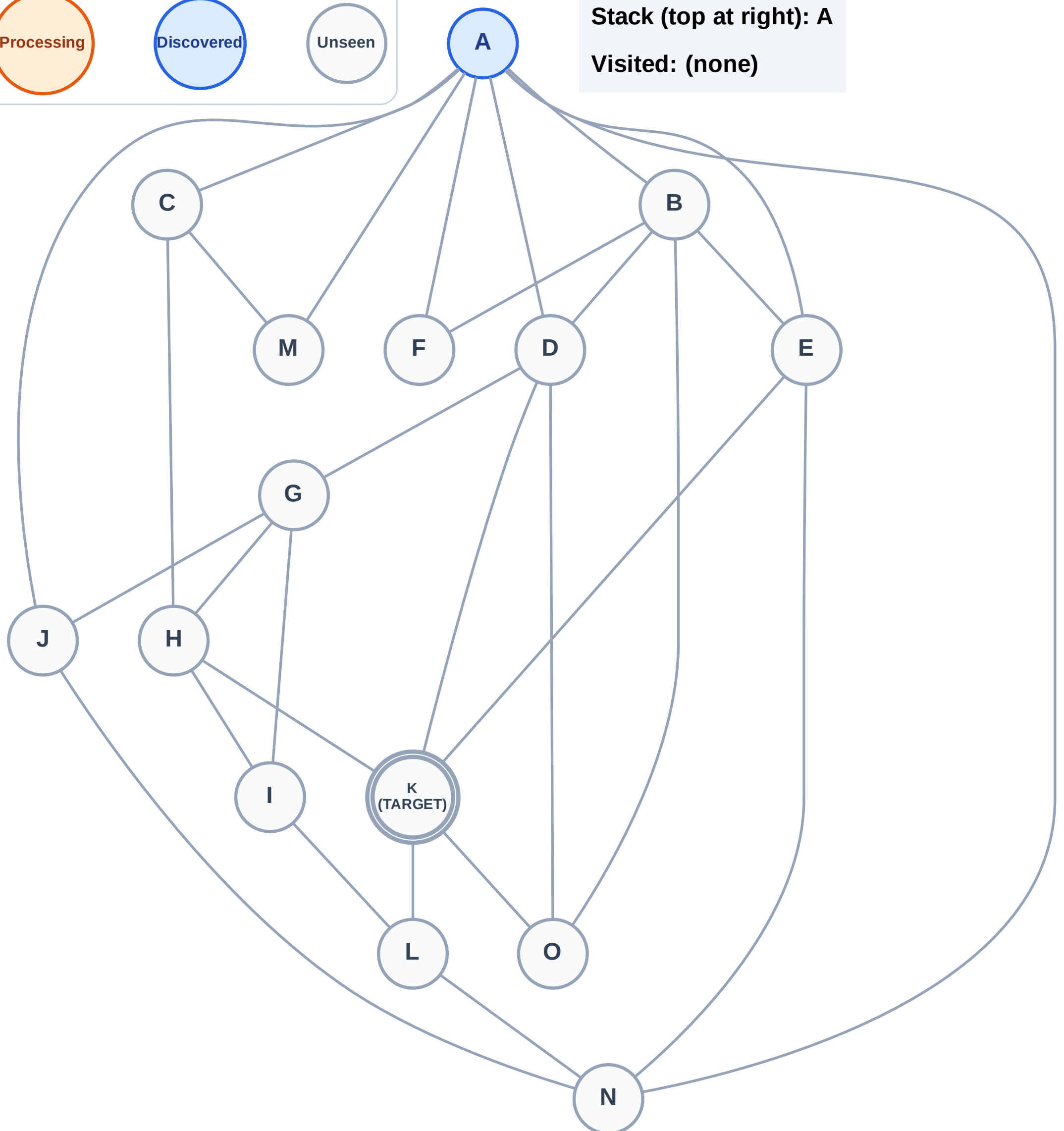
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

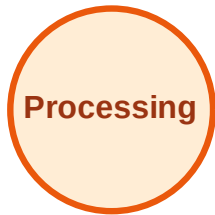
Visited: (none)



DFS Traversal

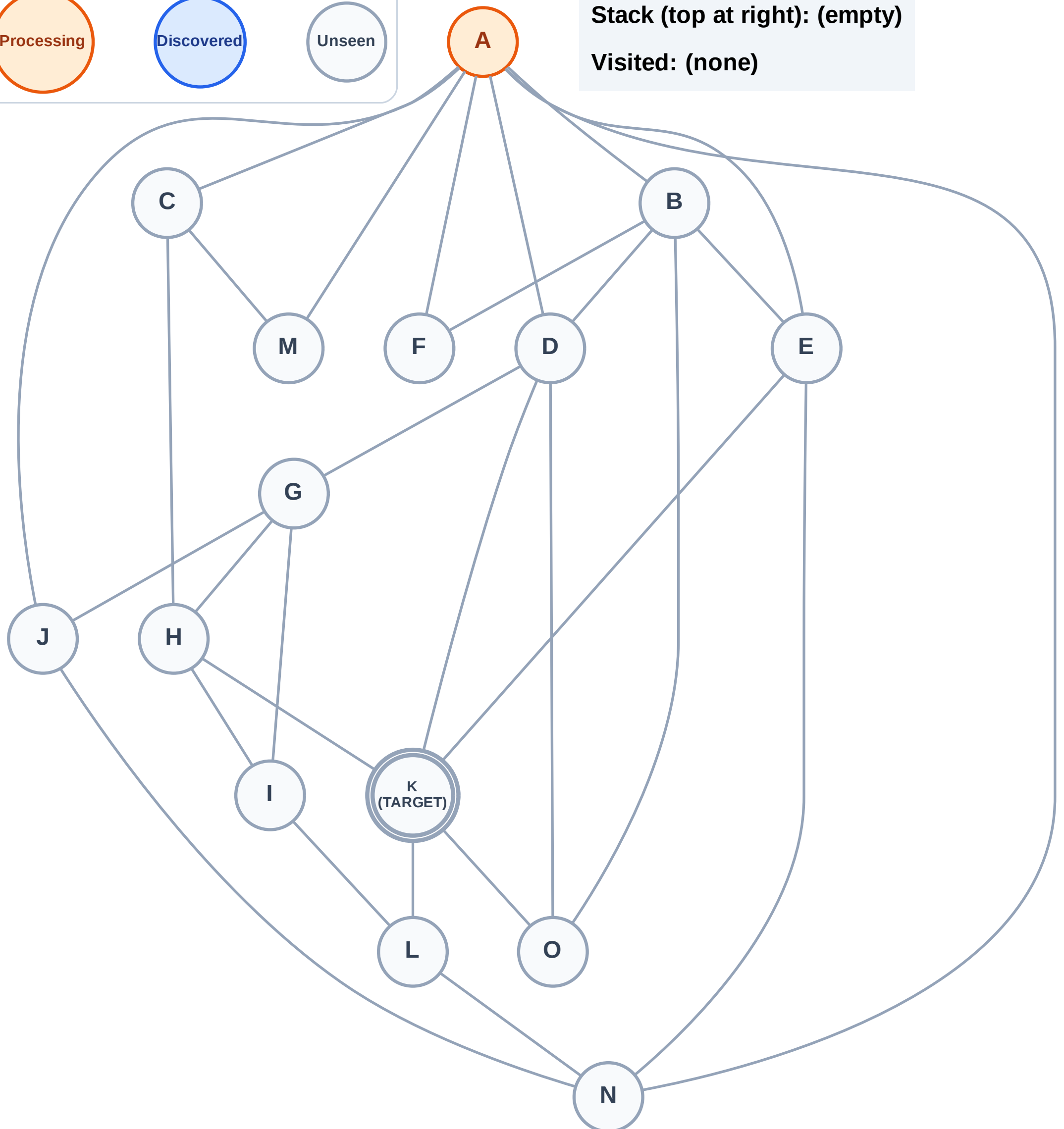
Step 2: Process node A

Legend



Stack (top at right): (empty)

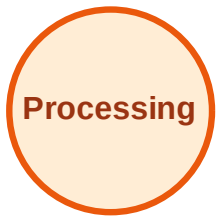
Visited: (none)



DFS Traversal

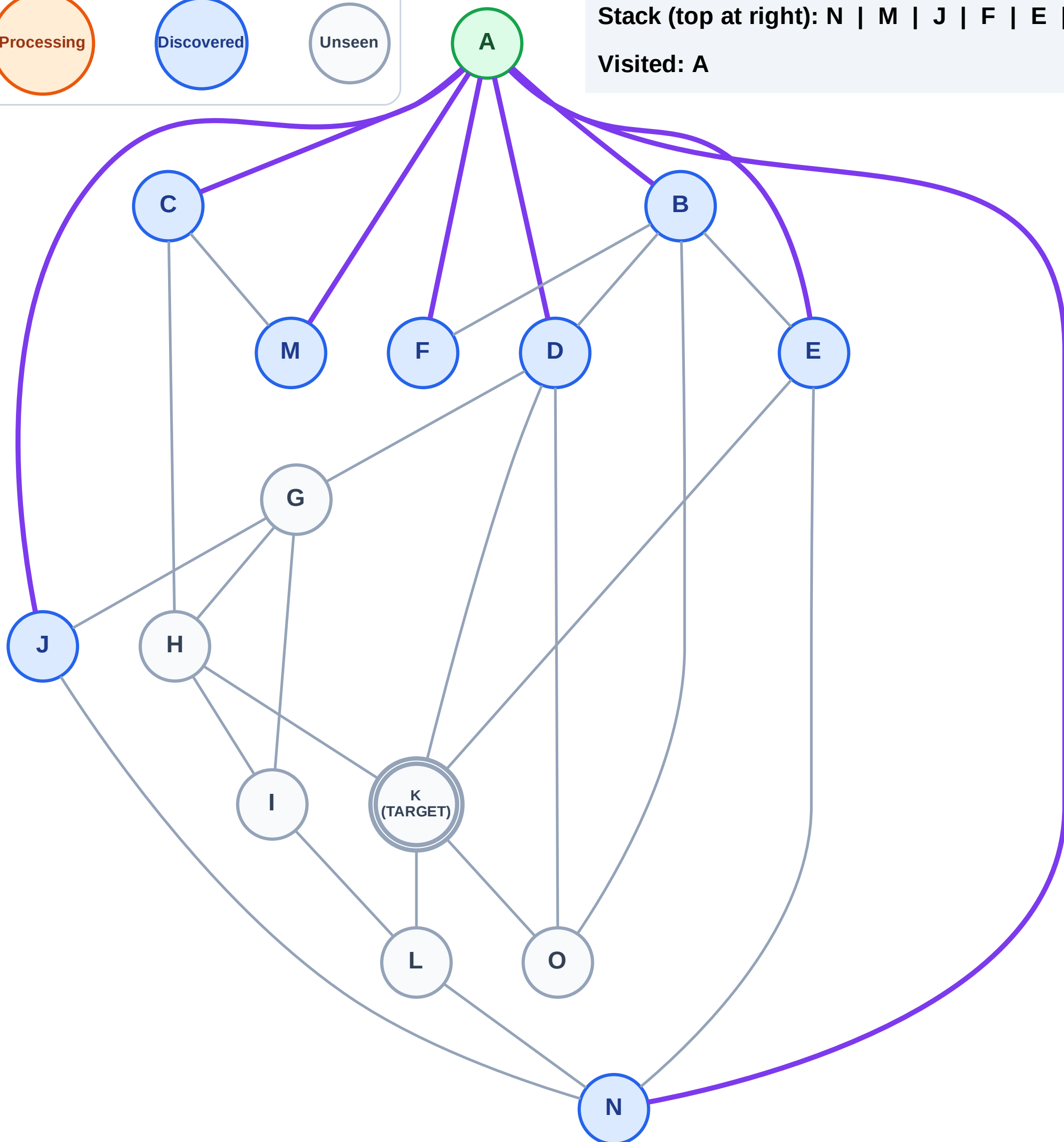
Step 3: Discover N, M, J, F, E, D, C, B from A

Legend



Stack (top at right): N | M | J | F | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

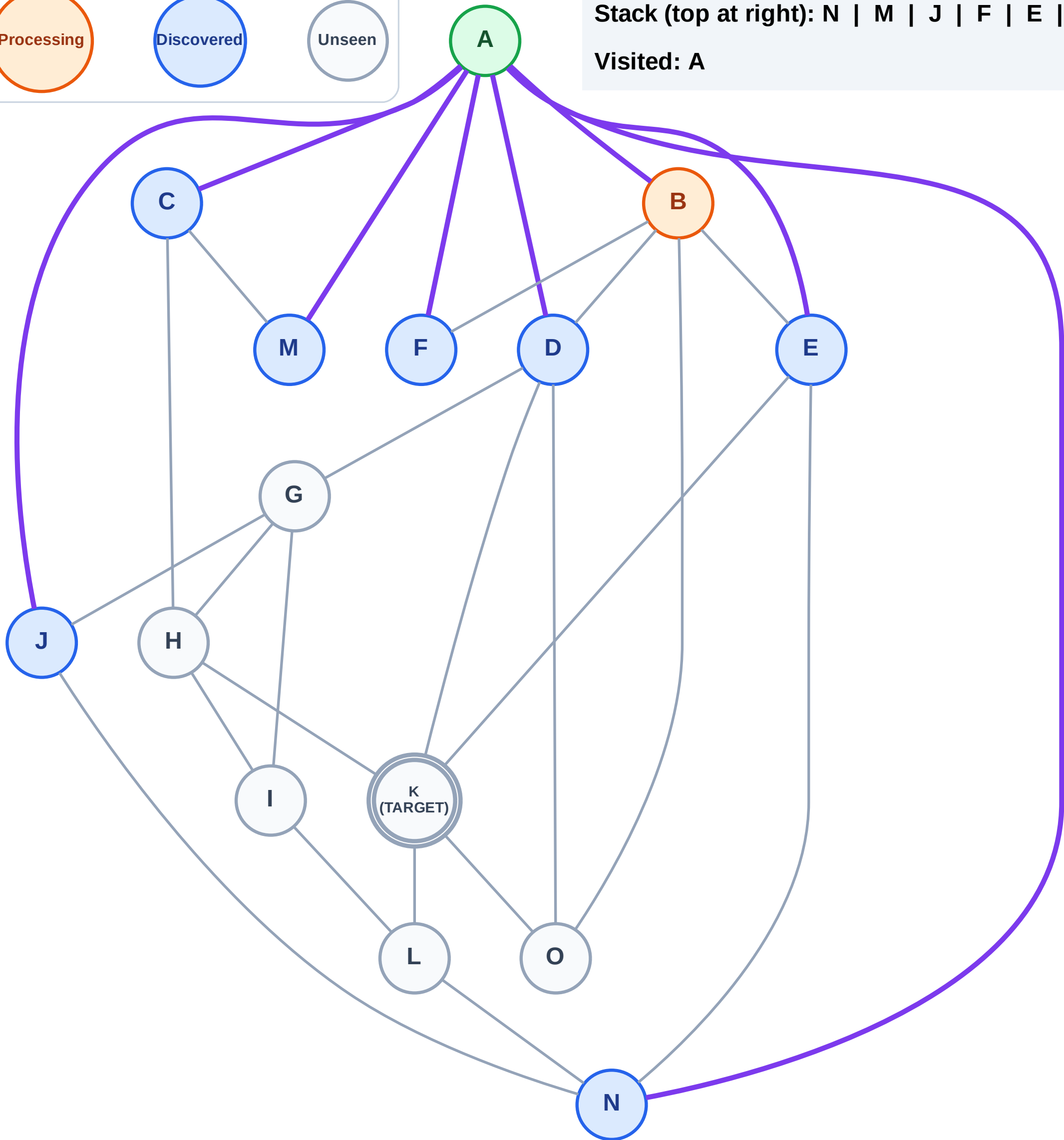
Processing

Discovered

Unseen

Stack (top at right): N | M | J | F | E | D | C

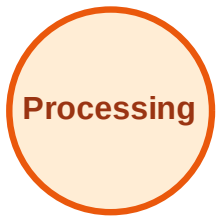
Visited: A



DFS Traversal

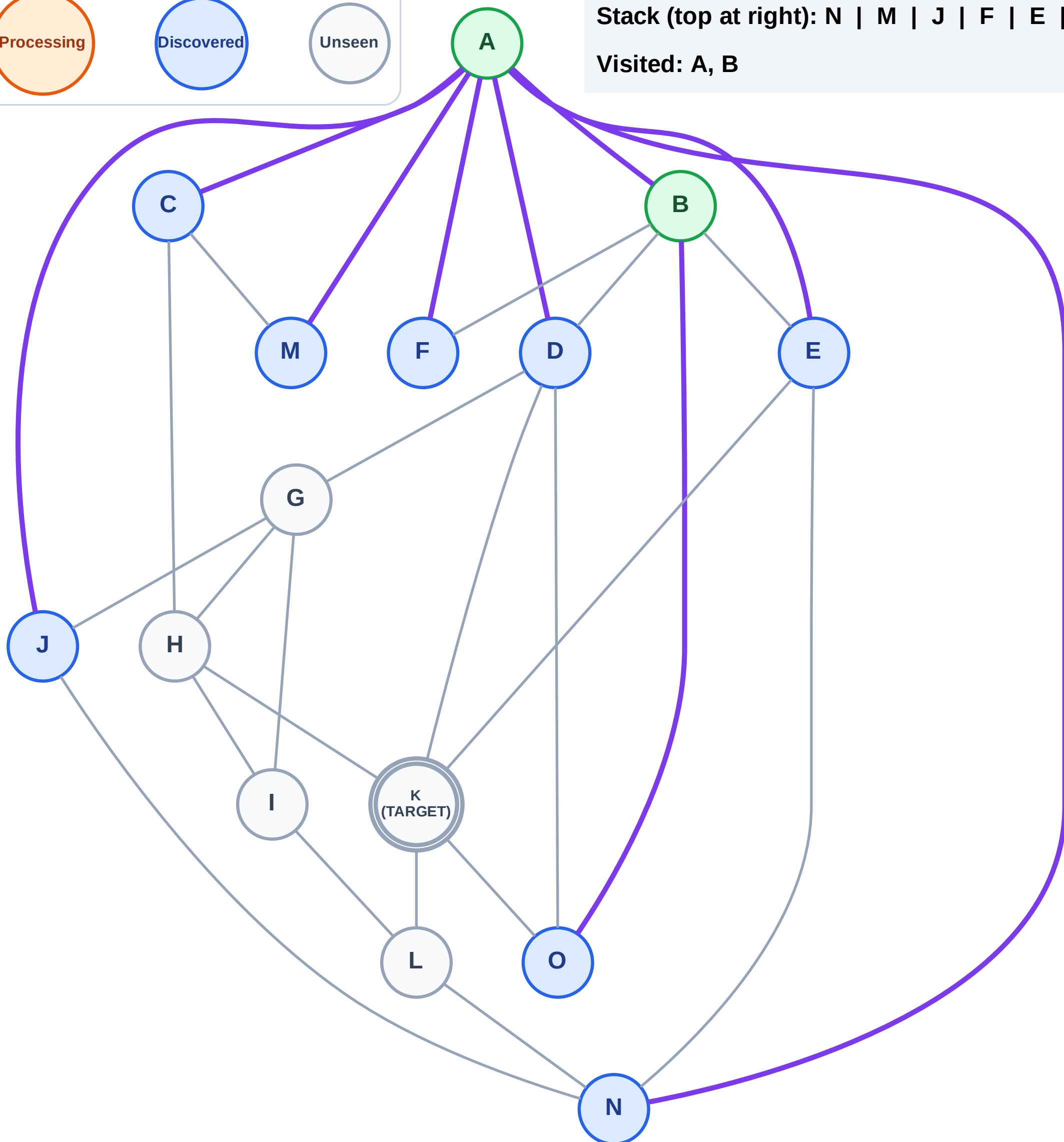
Step 5: Discover O from B

Legend



Stack (top at right): N | M | J | F | E | D | C | O

Visited: A, B



DFS Traversal

Step 6: Process node O

Legend

Complete

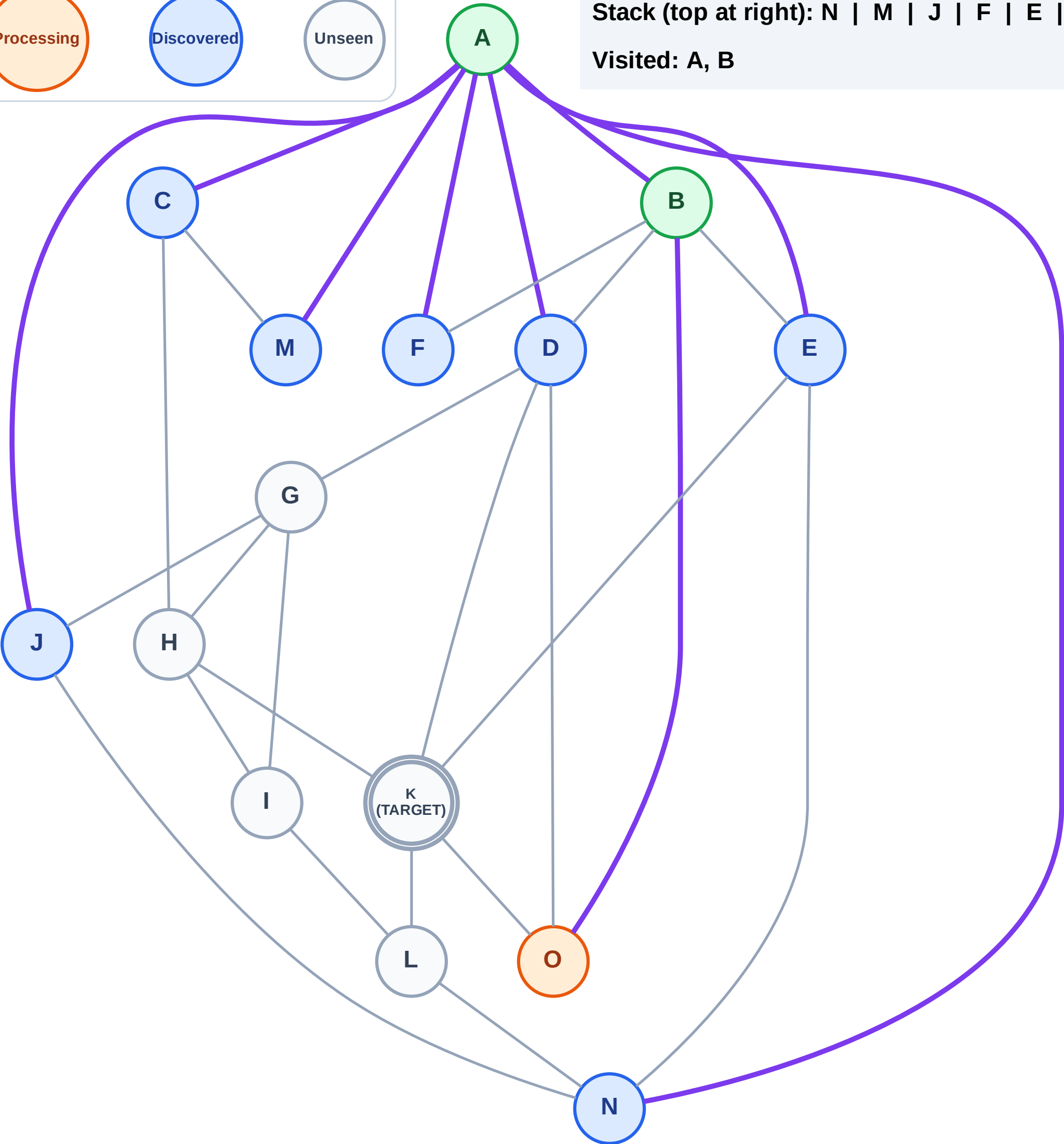
Processing

Discovered

Unseen

Stack (top at right): N | M | J | F | E | D | C

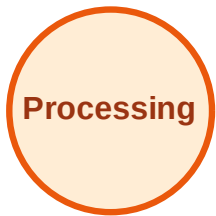
Visited: A, B



DFS Traversal

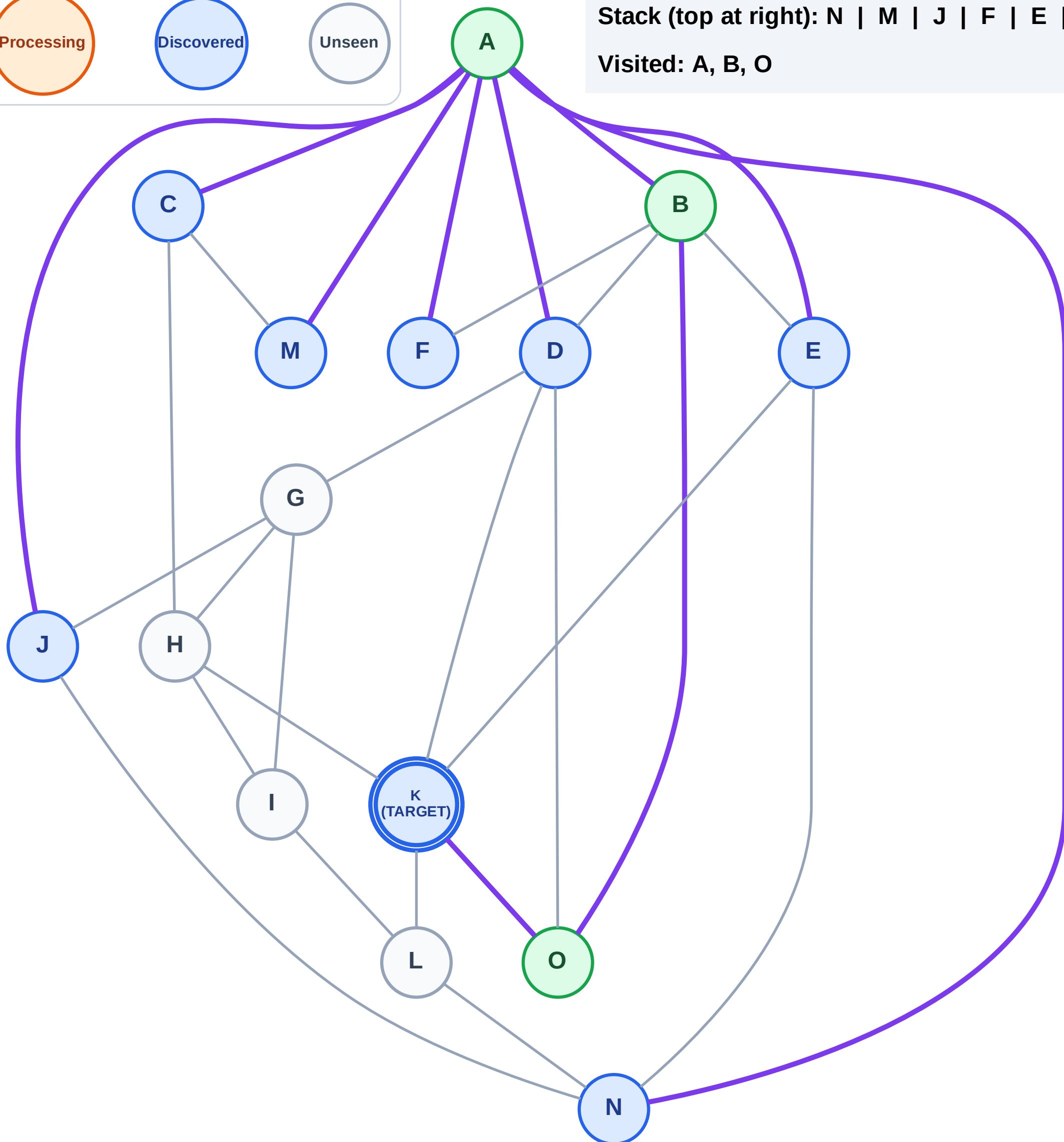
Step 7: Discover K from O

Legend



Stack (top at right): N | M | J | F | E | D | C | K

Visited: A, B, O



DFS Traversal

Step 8: Process node K

Legend

Complete

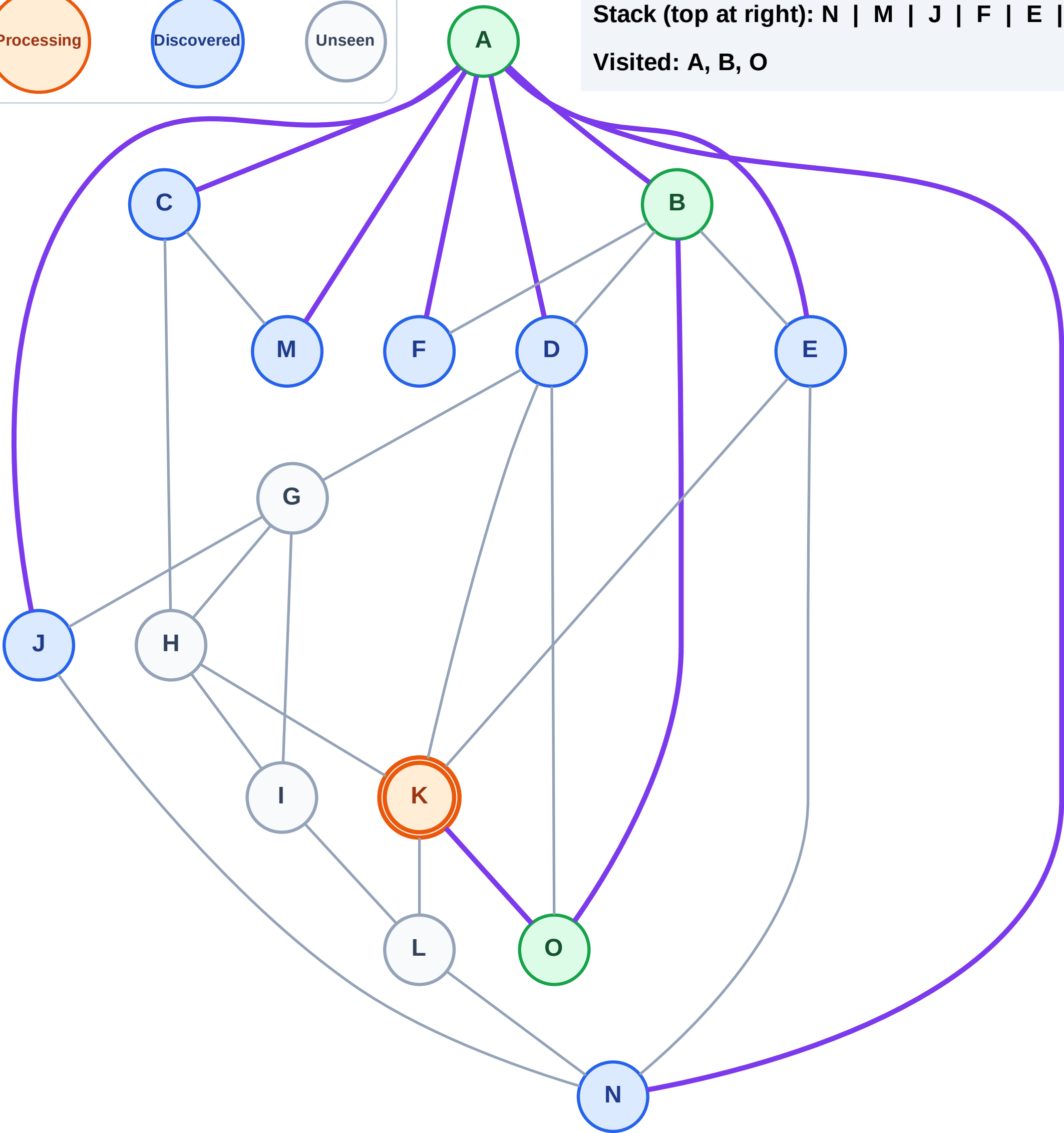
Processing

Discovered

Unseen

Stack (top at right): N | M | J | F | E | D | C

Visited: A, B, O



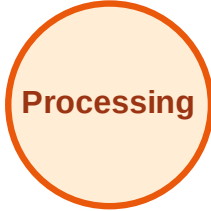
DFS Traversal

Step 9: TARGET NODE K REACHED! Path (3 hops): A → B → O → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): N | M | J | F | E | D | C

Visited: A, B, K, O

